Splunk server & Forwarder Installation

1) [Splunk Server] Splunk Installation

2) [CentOS8] Splunk Universal Forwarder installation

3) [Splunk Server] Forwarder 수집 환경 설정

4) [Win10] Splunk Universal Forwarder Installation

5) [Splunk Server] Sysmon 전용 프로그램 설치

Splunk 기능

"**머신데이터**를 아무런 **제약 없이** **수집>저장>분석>시각화** 할 수 있는 **실시간 분산** **플랫폼**"

머신데이터
(Machine Data)

- 서버/NW 로그
- 각종 설비 데이터
- 애플리케이션 로그
- 기타 **모든 텍스트 형태의 데이터**

제약 없음
(No Limits)

- 비정형/정형 데이터
- 데이터 **포맷 무관**
- 데이터 **용량 무관**
- 데이터 속도 무관
- 제약 없이 수용

엔드 투 엔드
(End-to-End)

- 별도의 외부 솔루션불필요
- 복잡한 코딩 및 SI 개발이 **필요 없음**
- 데이터의 생성부터 가치 획득까지 모두

실시간 및 분산
(Real-time)

- 모든 데이터 **실시간** 처리, 즉시 결과 확인
- **분산 저장, 분산 검색**
- 성능 및 용량의 선형적 확장

플랫폼
(Platform)

- 커스터마이징 용이
- 외부 시스템과 손쉬운 연동
- **1500여 무료 앱**을 통한 기능 확장
- 개발 프레임워크

* 머신 데이타

어플리케이션 데이터	모바일 앱 & 웹사이트 데이터
IT 인프라 데이터	네트워크 서버 & 클라우드 서비스
보안 데이터	방화벽 데이터 & 엔드포인트 보안 솔루션 데이터
고객 생성 데이터	소셜 미디어 데이터 & 콜 로그
IoT	온도 제어 & 속도 측정 장치 2017년 기준으로 장치(Thing)

• Splunk Platform

Splunk Cloud	• 평가판은 15일 동안 이용 가능 • 평가판 라이선스로 Splunk Cloud의 모든 기능과 프리미엄 애플리케이션 및 추가 기능을 사용 가능 • 매일 최대 5GB의 데이터를 인덱싱 가능
Splunk Enterprise	• 평가판은 60일 이용 가능 • 평가판 라이선스로 Splunk Enterprise의 모든 기능과 모든 프리미엄 애플리케이션 및 추가 기능을 사용 가능 • 매일 최대 500MB의 데이터를 인덱싱 가능 • 60일이 지나면 Enterprise 평가판 라이선스가 영구적인 Free 라이선스로 전환 - 사용자 기본 설정, 인증 및 경고 같은 일부 기능을 사용할 수 없음 • Free 라이선스도 인덱싱 볼륨이 매일 500MB로 제한되지만, 만료 날짜 없음

• 시스템 요구사항

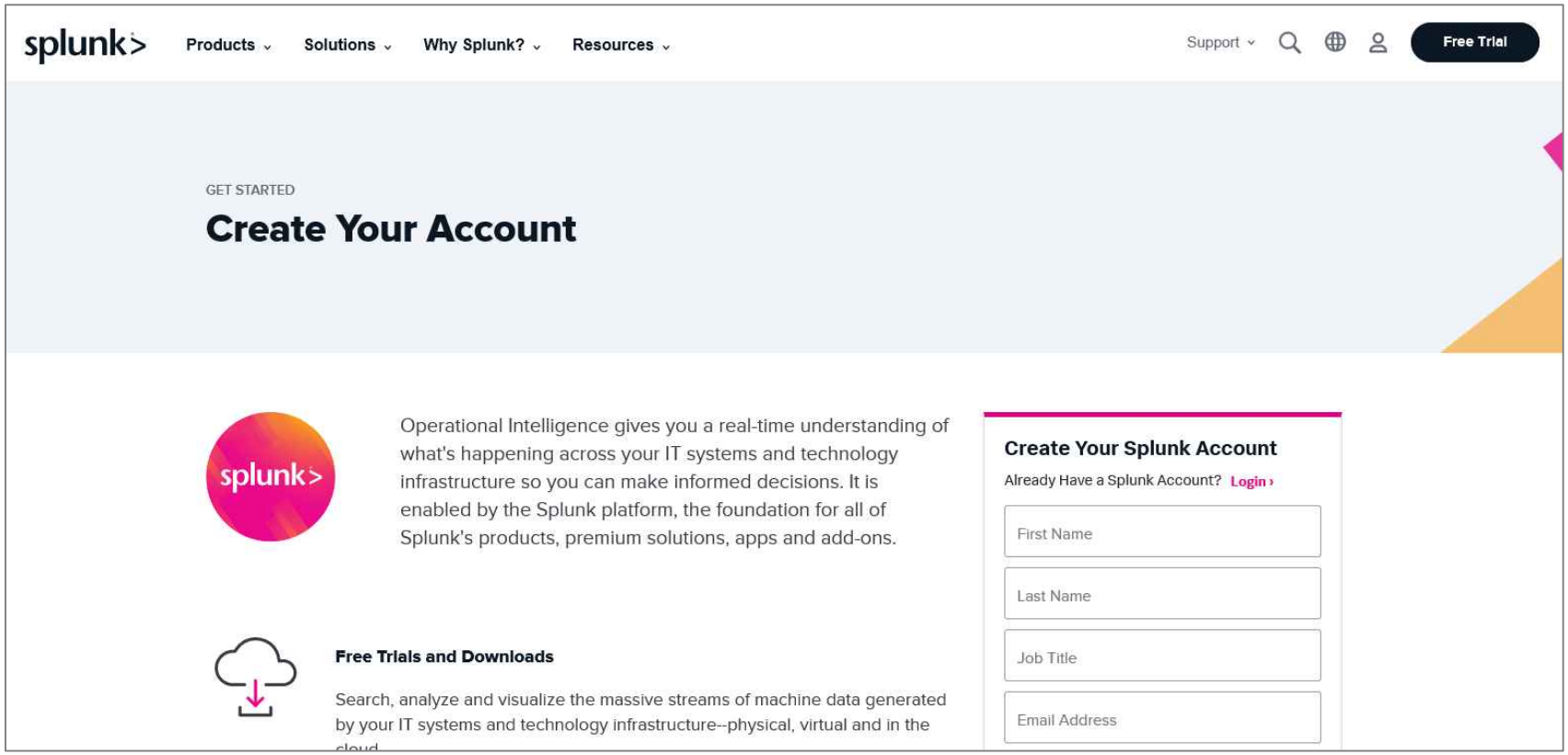
Splunk Cloud	• 웹 브라우저가 있어야 함 • Chrome, Firefox 및 Safari의 최신 버전에서 Splunk Cloud를 사용
Splunk Enterprise	• Linux, Windows 또는 Mac OS에서 사용가능 • Windows가 아닌 플랫폼 - 2코어 64비트 CPU 2GHz이상, 4GB RAM • Windows 플랫폼 - 2코어 64비트 CPU 2GHz이상, 4GB RAM • 웹브라우저 - Chrome, Firefox 및 Safari 브라우저의 최신 버전 지원

Windows 환경에서 Splunk 설치

- 계정 만들기
- 플랫폼 선택
- 시스템 요구사항
- 튜토리얼 데이터 파일 다운로드
- Splunk 소프트웨어 평가판 액세스

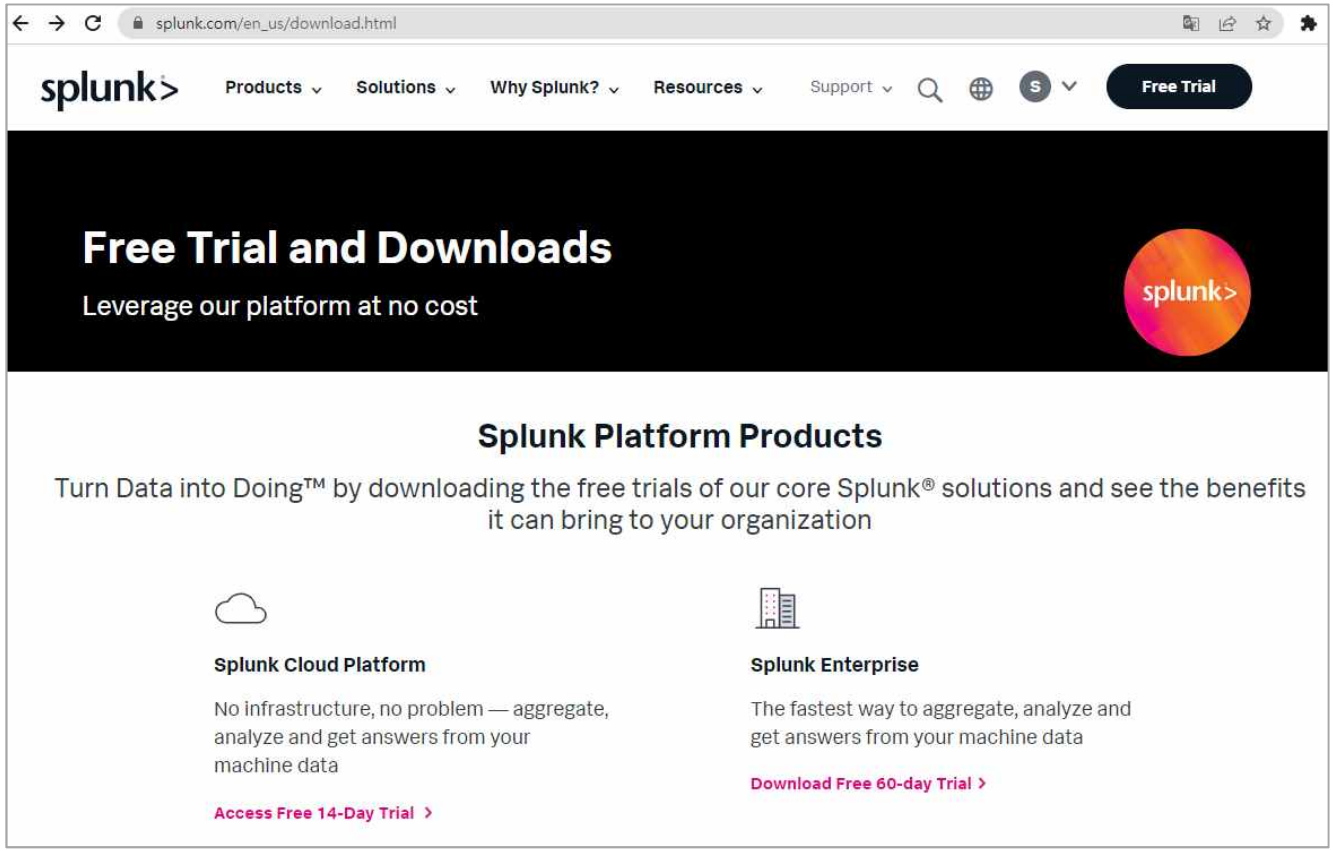
❶ 계정 만들기

http://www.splunk.com > 클릭



❷ Splunk 소프트웨어 평가판 Download

- Splunk 웹 사이트 https://www.splunk.com
- 오른쪽 상단 모서리에서 **Free Splunk(Free trial)** 클릭
- 사용할 플랫폼을 선택하고 링크를 클릭하여 평가판 소프트웨어 다운로드



splunk> Products Solutions Why Splunk? Resources Support Free Trial

GET STARTED

Choose Your Download

Splunk Enterprise 9.0.0

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

Windows Linux Mac OS

64-bit	Windows 10 Windows Server 2016, 2019, 2022	.msi	454.15 MB	Download Now

splunk> Products Solutions Why Splunk? Resources Support S Free Trial

SPLUNK GENERAL TERMS

Last Updated: August 12, 2021

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Access program

❸ Tutorial 데이터 파일 다운로드

- 튜토리얼에서 사용할 데이터 파일 다운로드

  - CSV 파일 형식인 데이터 파일에는 웹 액세스 로그 파일, 보안 형식 로그 파일, 판매 로그 파일 및 가격 리스트가 포함되어 있음

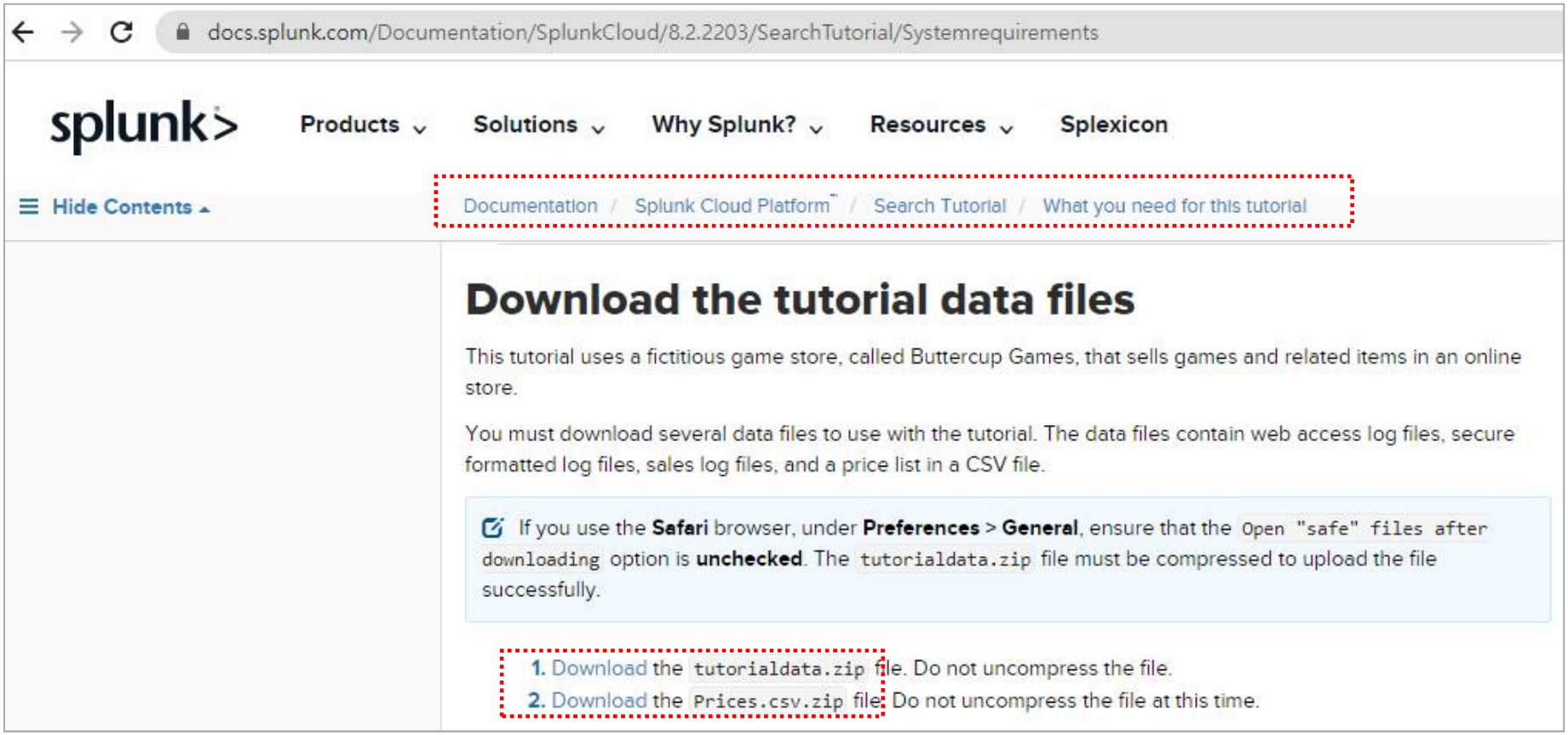
- tutorialdata.zip 파일을 성공적으로 업로드하려면 압축해야 함

  - tutorialdata.zip 파일 다운로드(파일 압축해제 하지 않음)

  - Prices.csv.zip 파일 다운로드(파일 압축해제 하지 않음)

- **Tutorial 데이터 파일 다운로드 사이트**

https://docs.splunk.com/Documentation/SplunkCloud/8.2.2203/SearchTutorial/Systemrequirements

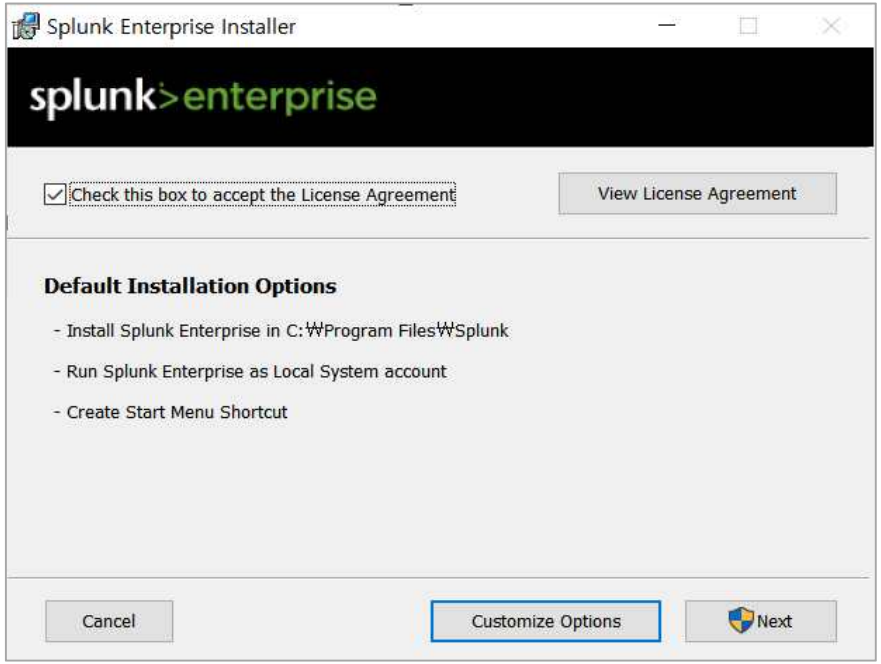


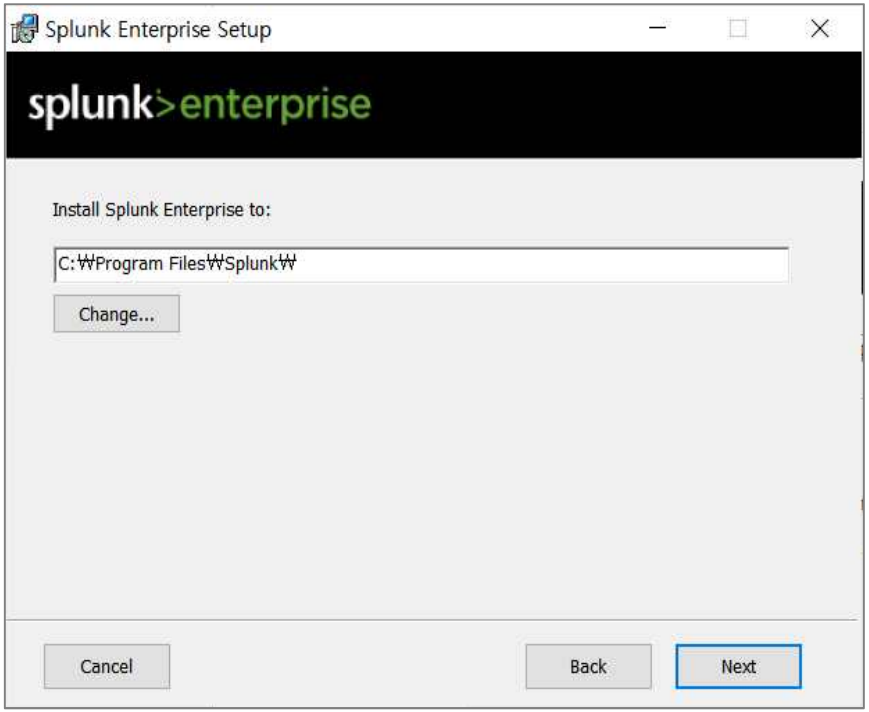
❹ Splunk Enterprise 설치

- 설치 프로그램이 있는 폴더나 디렉터리로 이동
- splunk.msi 파일을 두 번 클릭하여 설치 프로그램을 시작
  - **'라이선스 계약에 동의합니다'** 클릭 > **다음** 클릭
  - 사용자 ID와 암호를 지정(암호 길이는 8자 이상)
    * 입력하는 동안에는 커서가 이동하지 않음
- 설치 완료 패널에서 **Splunk로 브라우저 실행** 체크박스가 선택되어 있는지 확인

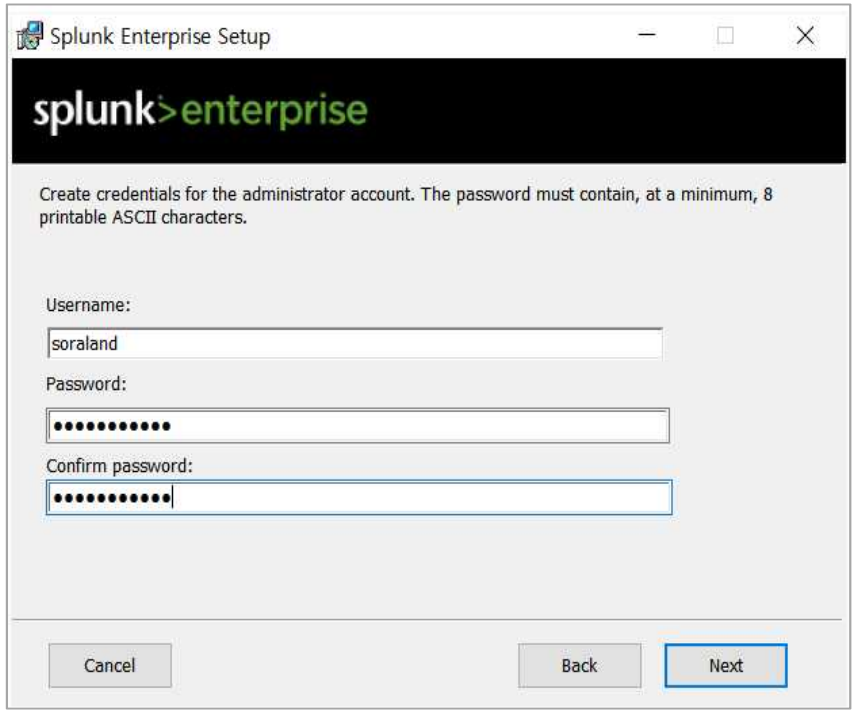
  마침 클릭
- Splunk Web 실행 절차로 이동

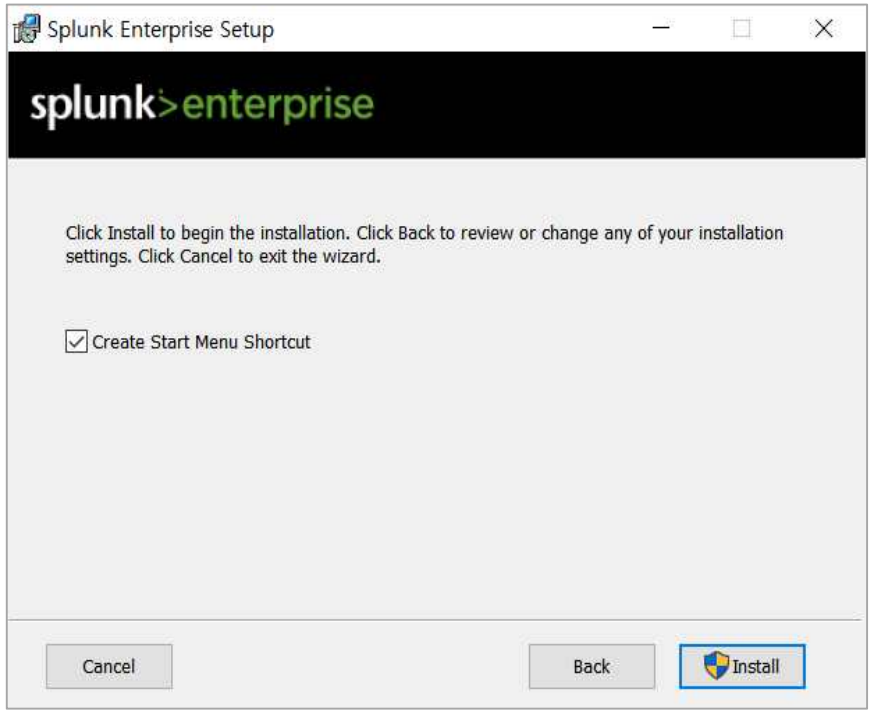
Splunk Enterprise Setup
splunk> enterprise
Welcome to the Splunk Enterprise Setup Wizard
Please wait while the Setup Wizard prepares to guide you through the installation or upgrade.
Computing space requirements
Back
Next
Cancel

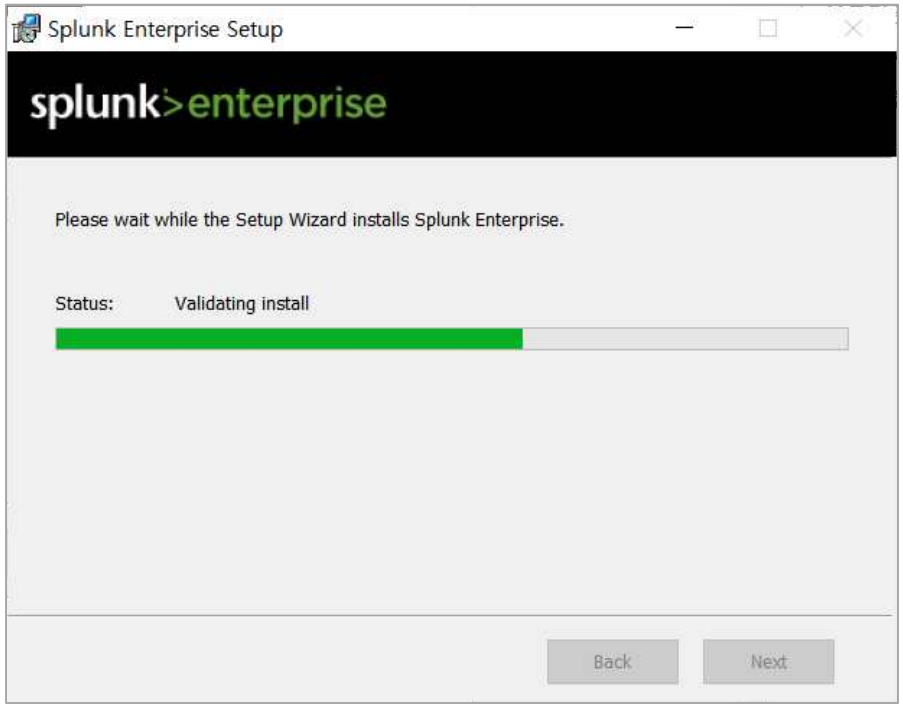

Splunk Enterprise Installer
splunk>enterprise
Check this box to accept the License Agreement
View License Agreement
Default Installation Options
- Install Splunk Enterprise in C:₩Program Files₩Splunk
- Run Splunk Enterprise as Local System account
- Create Start Menu Shortcut
Cancel
Customize Options
Next

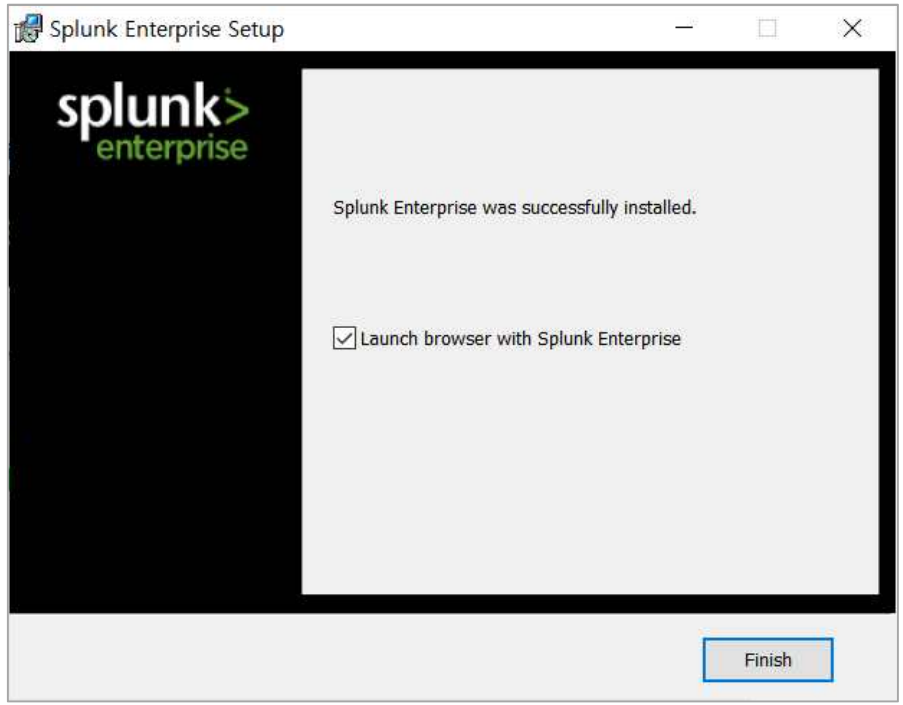

Splunk Enterprise Setup
splunk>enterprise
Install Splunk Enterprise to:
C:₩Program Files₩Splunk₩
Change...
Cancel
Back
Next

Splunk Enterprise Setup
splunk>enterprise
The user you install Splunk Enterprise as determines what data it has access to
Install Splunk Enterprise as
Local System
Installs Splunk Enterprise using local system account. Splunk Enterprise can access all data on or forwarded to this machine.
Domain Account
Installs Splunk Enterprise with domain account you provide. This lets you collect logs and metrics from remote machines as well as local and forwarded data. You can set the account in the next dialog.
Cancel
Back
Next

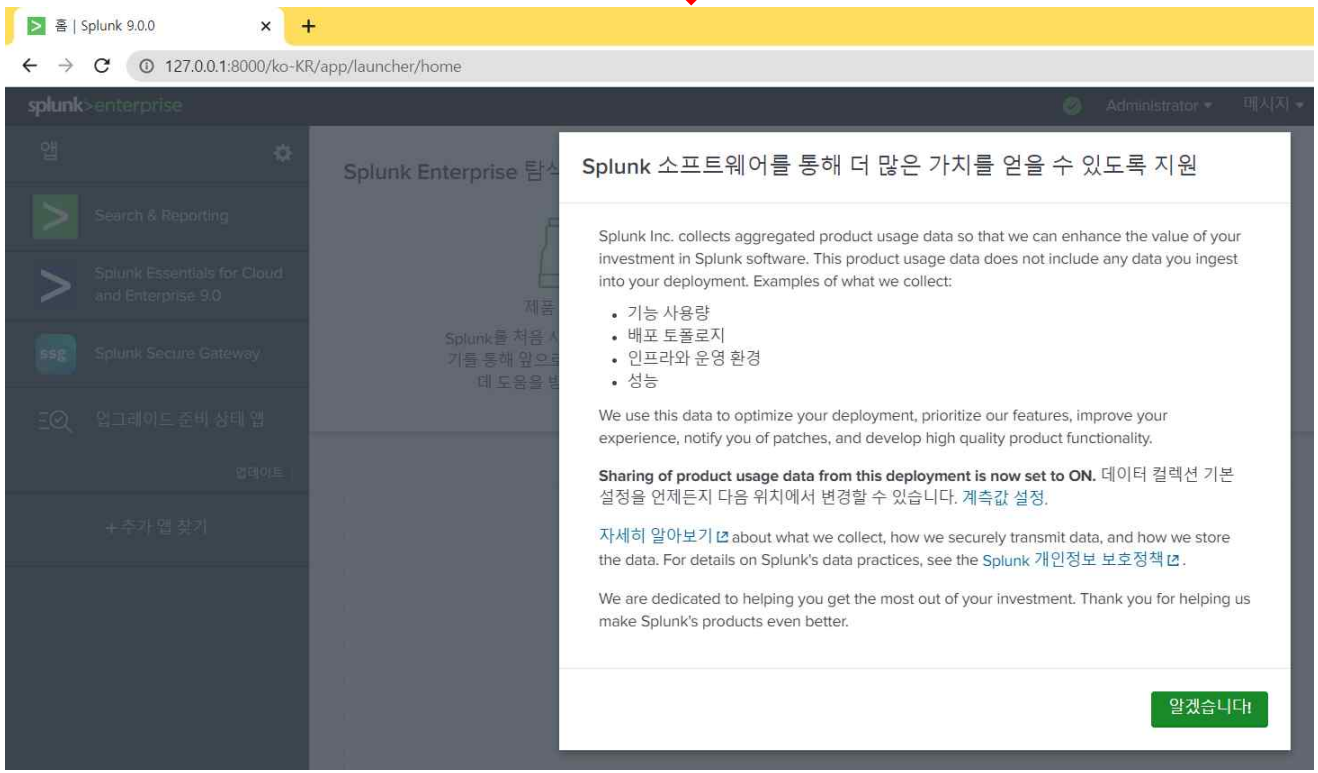

Splunk Enterprise Setup
splunk>enterprise
Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.
Username:
soraland
Password:
Confirm password:
Cancel
Back
Next


Splunk Enterprise Setup
splunk>enterprise
Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.
Create Start Menu Shortcut
Cancel
Back
Install

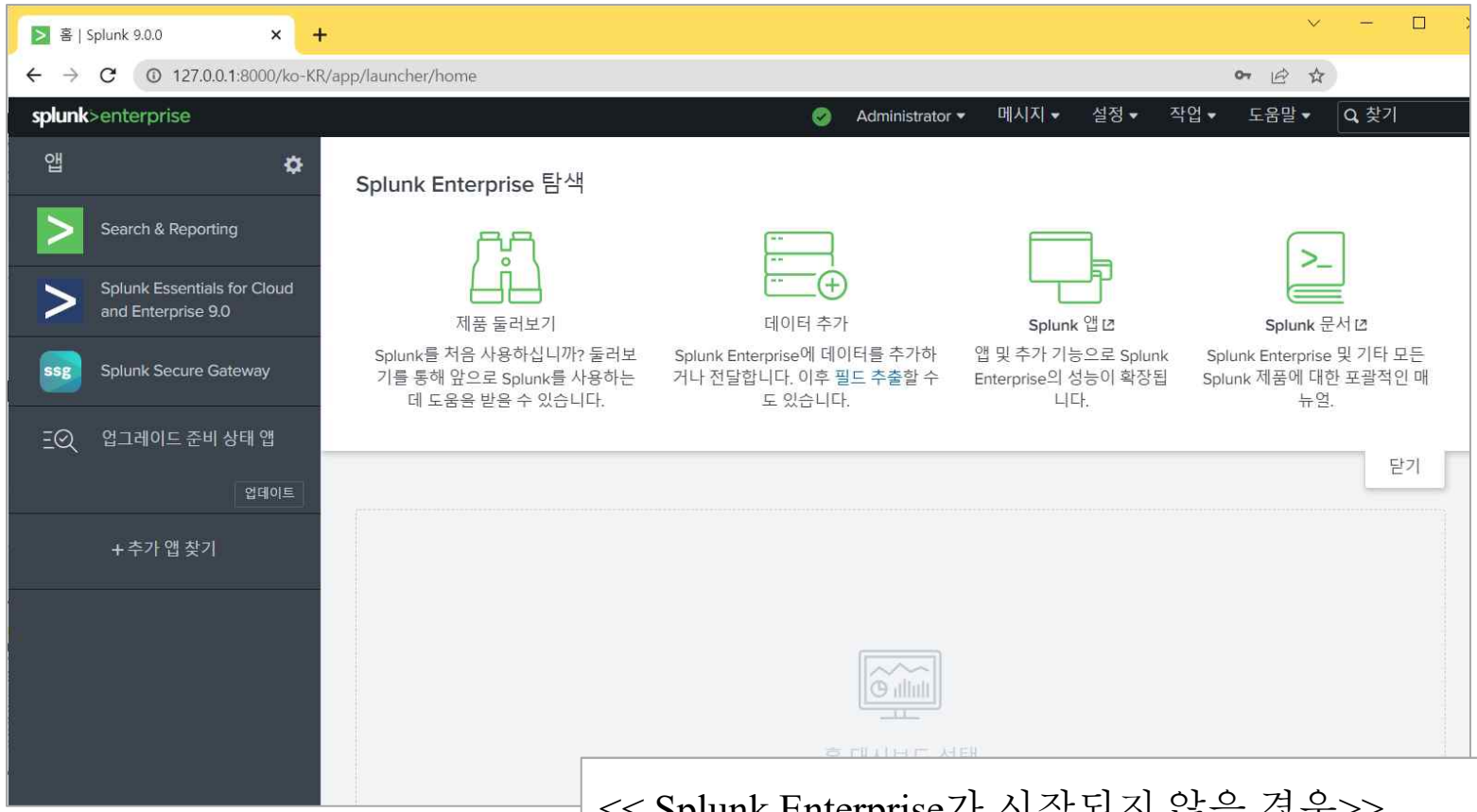

Splunk Enterprise Setup
splunk>enterprise
Please wait while the Setup Wizard installs Splunk Enterprise.
Status:
Validating install
Back
Next


Splunk Enterprise Setup
splunk>
enterprise
Splunk Enterprise was successfully installed.
Launch browser with Splunk Enterprise
Finish

http://localhost:8000
또는
http://127.0.0.1:8000

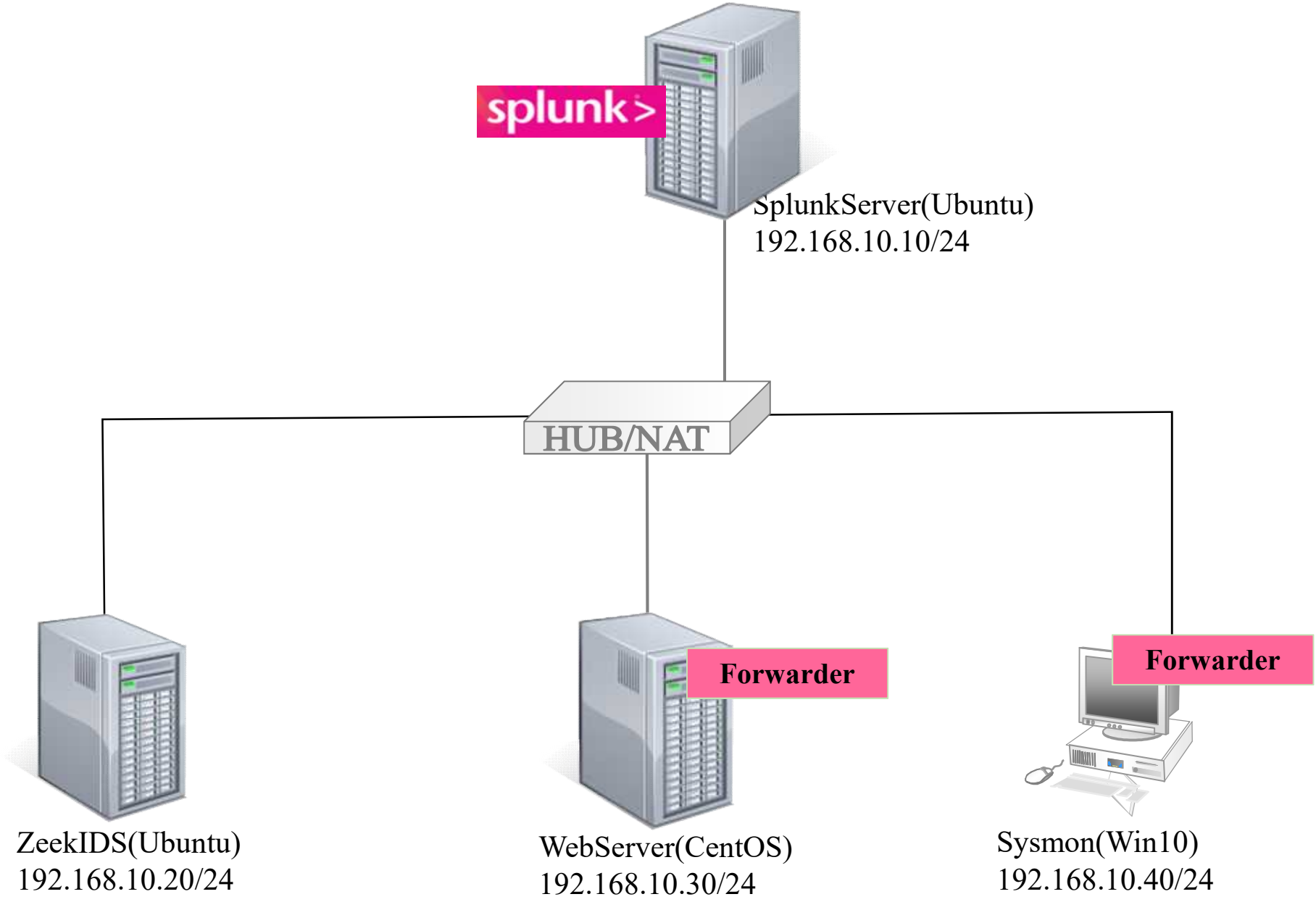


❺ Splunk Web 실행

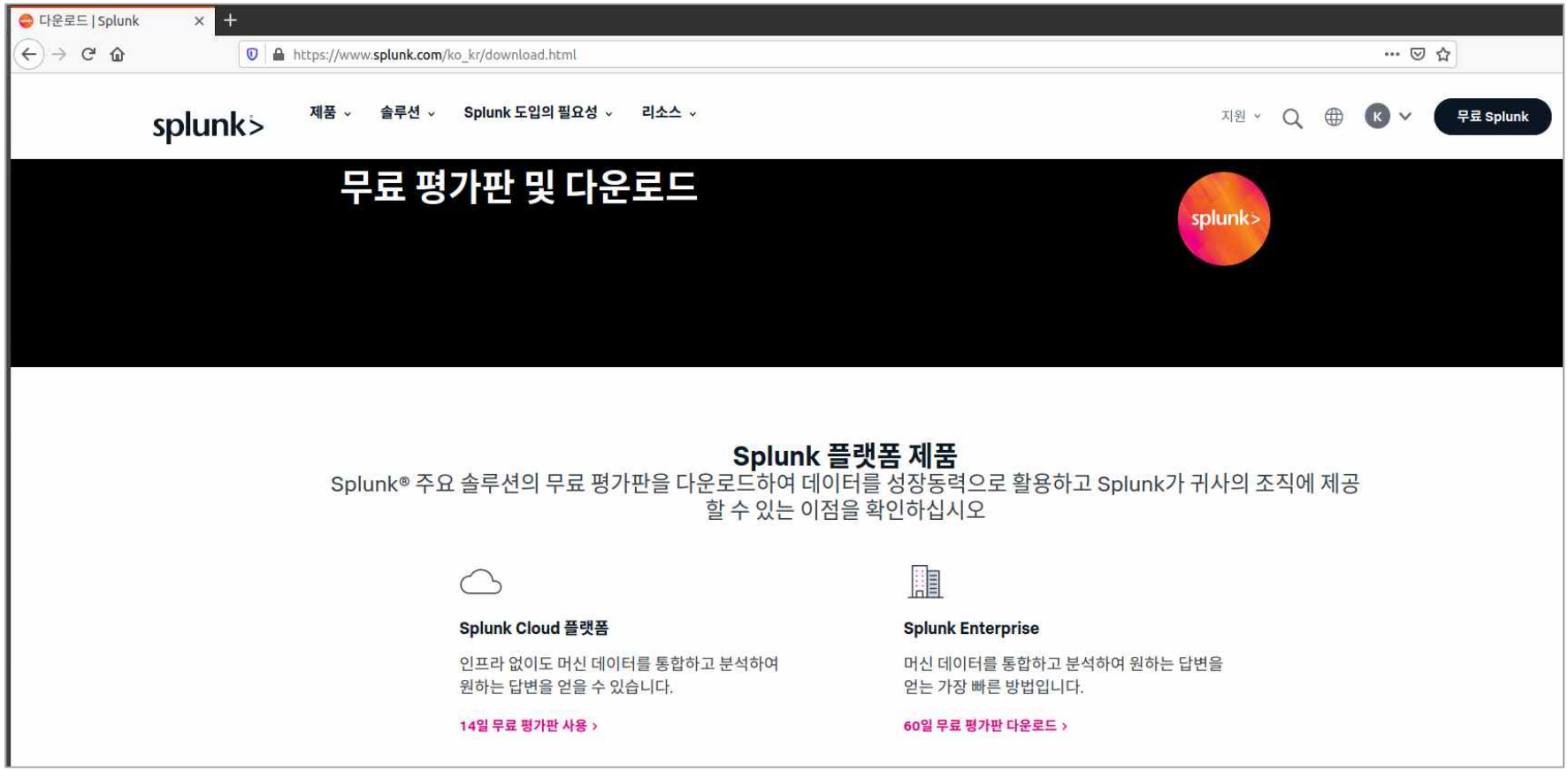


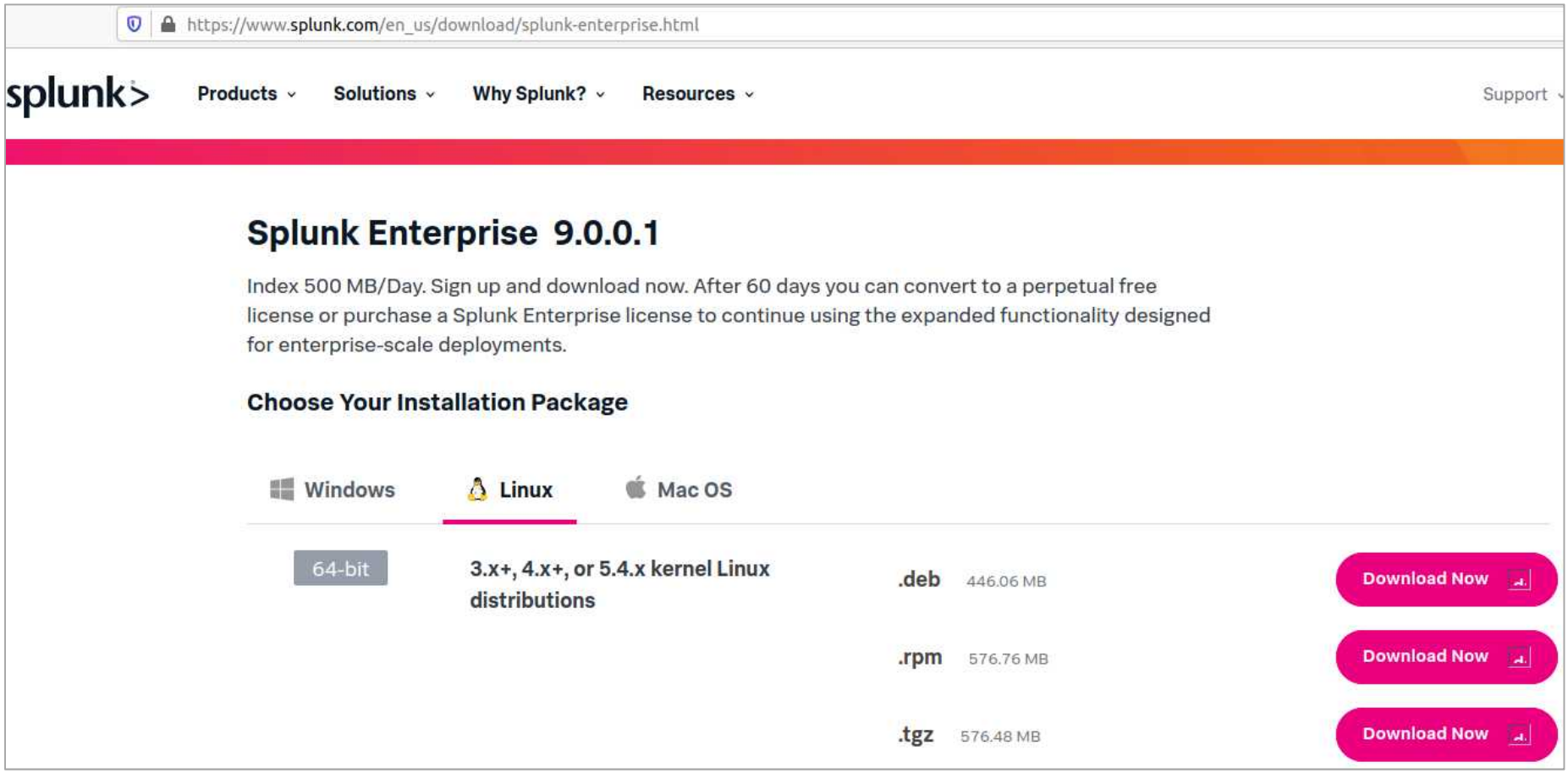
<< Splunk Enterprise가 시작되지 않을 경우>>

- 시작 > Splunk Enterprise 시작
- Windows 서비스 관리자 > Splunk Enterprise를 시작
- **CMD > \Program Files\Splunk\bin > splunk start**


splunk>
SplunkServer(Ubuntu)
192.168.10.10/24
HUB/NAT
Forwarder
Forwarder
ZeekIDS(Ubuntu)
192.168.10.20/24
WebServer(CentOS)
192.168.10.30/24
Sysmon(Win10)
192.168.10.40/24

1) [Splunk Server] Splunk Installation




https://www.splunk.com/en_us/download/splunk-enterprise.html
splunk>
Products
Solutions
Why Splunk?
Resources
Support
Splunk Enterprise 9.0.0.1
Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.
Choose Your Installation Package
Windows
Linux
Mac OS
64-bit
3.x+, 4.x+, or 5.4.x kernel Linux distributions
.deb 446.06 MB
Download Now
.rpm 576.76 MB
Download Now
.tgz 576.48 MB
Download Now

```
$cd /home/splunk/다운로드

$pwd

$ls –l

$splunk-9.0.0.1-9e907cedecb1-linux-2.6-amd64.deb

#su –

#apt –y install curl

#dpkg -i splunk-9.0.0.1-9e907cedecb1-linux-2.6-amd64.deb
```

#cd /opt/splunk/bin

#./splunk enable boot-start

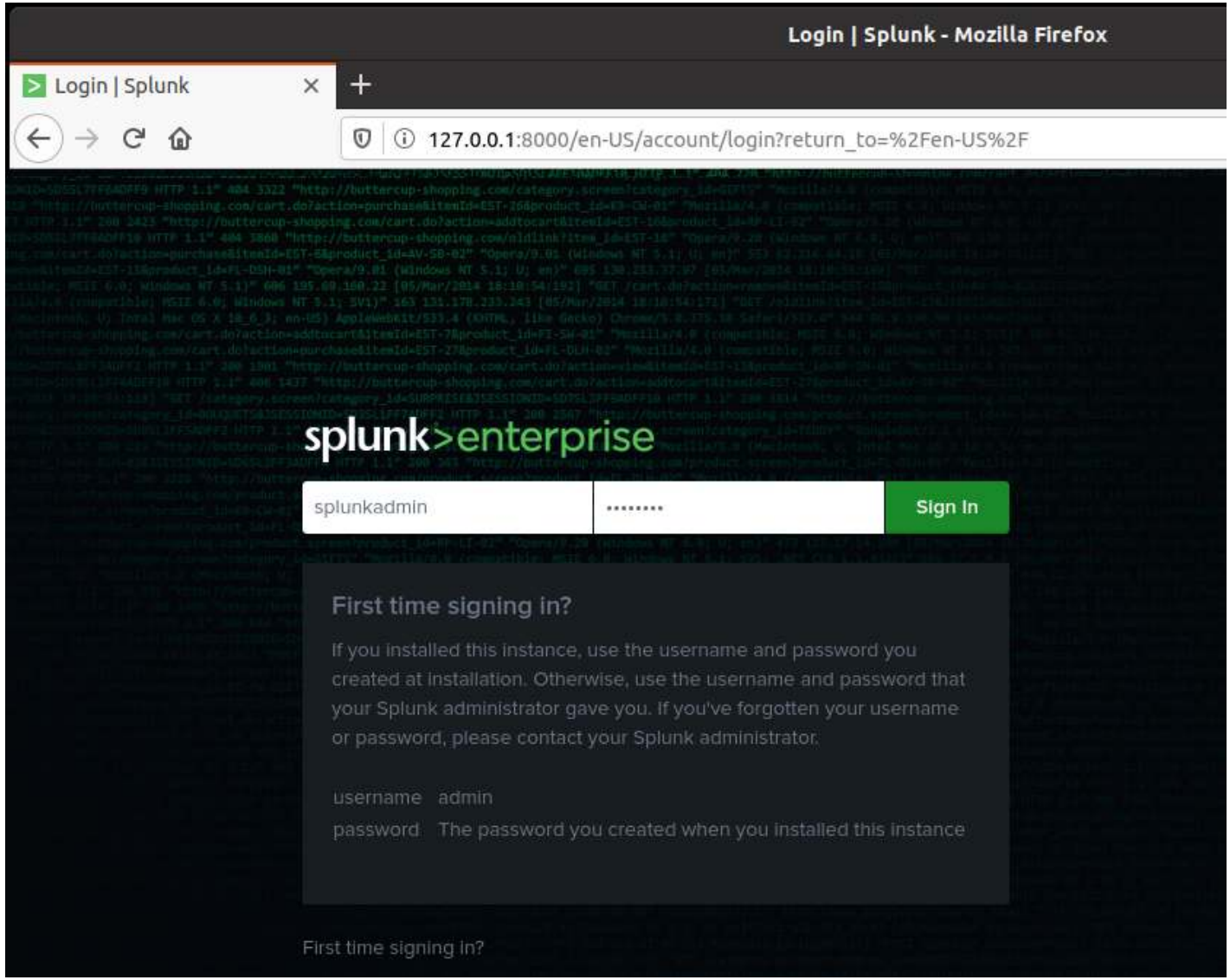
```
"Statement of Work" means the statements of work and/or any and all applicable
Orders, that describe the specific services to be performed by Splunk,
including any materials and deliverables to be delivered by Splunk.
Do you agree with this license? [y/n]: y

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: splunkadmin
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Generating RSA private key, 2048 bit long modulus
.....................................................................+++++
.....................................................................................................
.....................................................................+++++
e is 65537 (0x10001)
writing RSA key

Generating RSA private key, 2048 bit long modulus
.....................................................................+++++
..........+++++
e is 65537 (0x10001)
writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
root@splunk-server:/opt/splunk/bin#
```

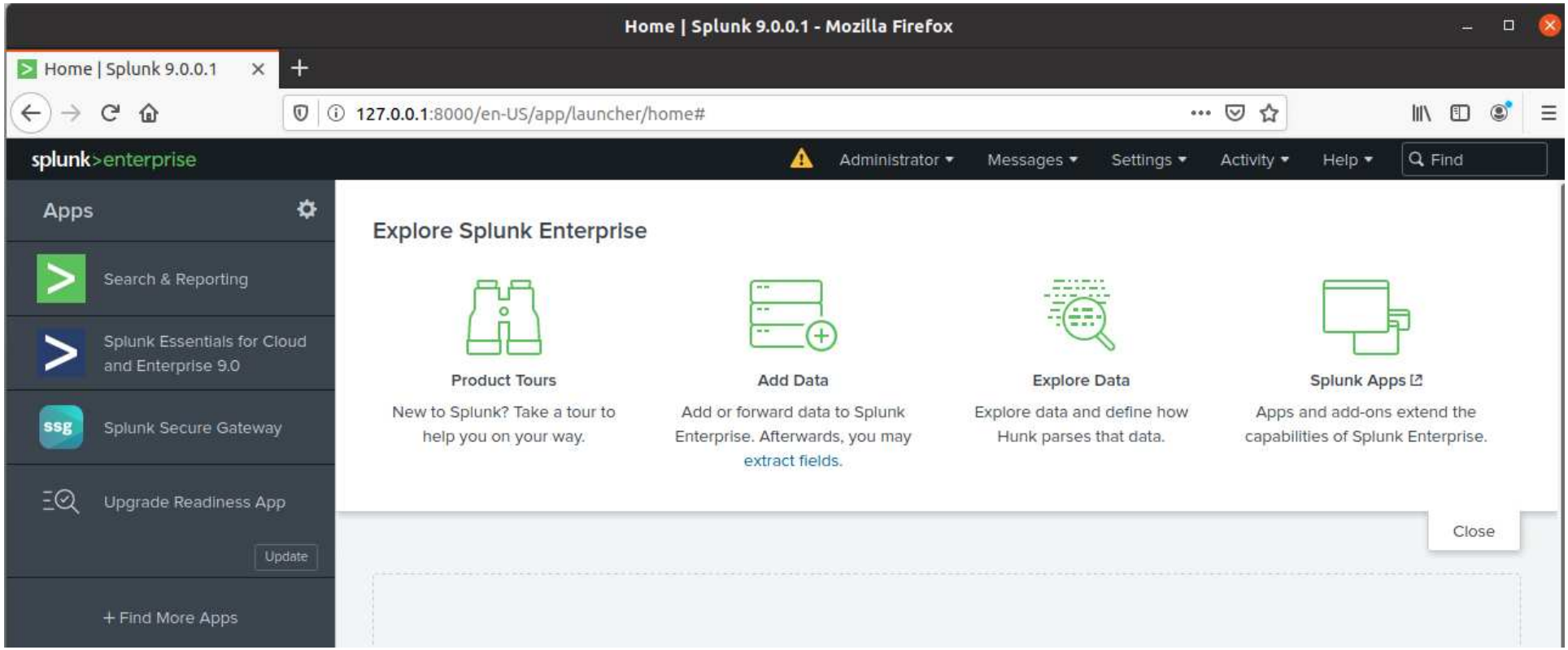
#cd /opt/splunk/bin

#./splunk start

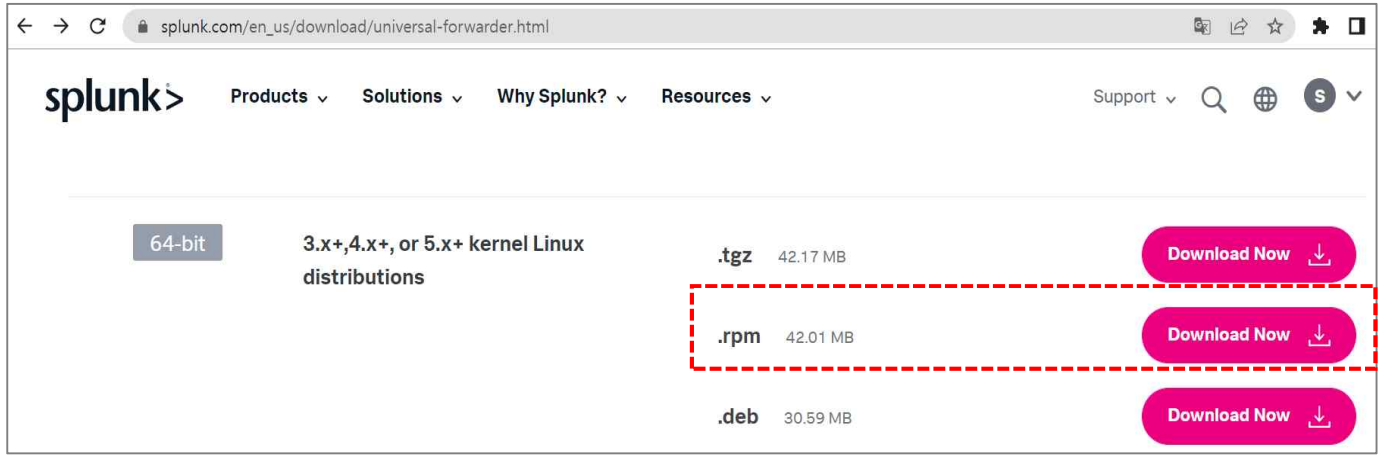
#./splunk status

```
root@splunk-server:/opt/splunk/bin#
root@splunk-server:/opt/splunk/bin# pwd
/opt/splunk/bin
root@splunk-server:/opt/splunk/bin# ./splunk status
splunkd is running (PID: 1420).
splunk helpers are running (PIDs: 1421 2209 2401 2520 5841).
root@splunk-server:/opt/splunk/bin#
```

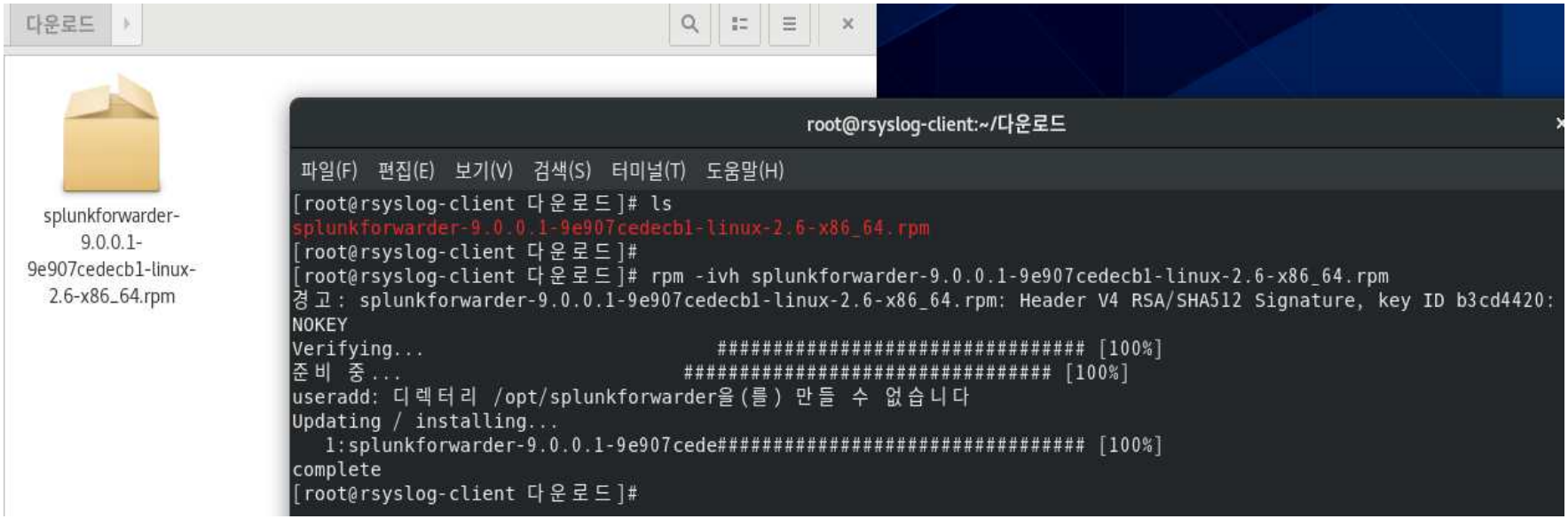

Login | Splunk - Mozilla Firefox
Login | Splunk
127.0.0.1:8000/en-US/account/login?return_to=%2Fen-US%2F
splunk>enterprise
splunkadmin
Sign In
First time signing in?
If you installed this instance, use the username and password you created at installation. Otherwise, use the username and password that your Splunk administrator gave you. If you've forgotten your username or password, please contact your Splunk administrator.
username admin
password The password you created when you installed this instance
First time signing in?


Home | Splunk 9.0.0.1 - Mozilla Firefox
Home | Splunk 9.0.0.1
127.0.0.1:8000/en-US/app/launcher/home#
splunk>enterprise
Administrator
Messages
Settings
Activity
Help
Find
Apps
Search & Reporting
Splunk Essentials for Cloud and Enterprise 9.0
Splunk Secure Gateway
Upgrade Readiness App
Update
+ Find More Apps
Explore Splunk Enterprise
Product Tours
New to Splunk? Take a tour to help you on your way.
Add Data
Add or forward data to Splunk Enterprise. Afterwards, you may extract fields.
Explore Data
Explore data and define how Hunk parses that data.
Splunk Apps
Apps and add-ons extend the capabilities of Splunk Enterprise.
Close

2) [CentOS8] Splunk Universal Forwarder installation



#rpm –ivh splunkforwarder-9.XXXX~~X86_64.rpm



- Splunkforwarder 설치 위치

  /opt/splunkforwarder/etc/apps/SplunkUniversalForwarder

- Splunkforwarder 환경설정 디렉터리

  /opt/splunkforwarder/etc/apps/SplunkUniversalForwarder/default

```
[root@rsyslog-client SplunkUniversalForwarder]# pwd
/opt/splunkforwarder/etc/apps/SplunkUniversalForwarder
[root@rsyslog-client SplunkUniversalForwarder]# ls -l
합계 0
drwxr-xr-x. 2 splunk splunk 191  8월  7 05:14 default
drwxr-xr-x. 2 splunk splunk  26  8월  7 05:14 metadata
[root@rsyslog-client SplunkUniversalForwarder]# mkdir local
[root@rsyslog-client SplunkUniversalForwarder]# cp default/* local/
[root@rsyslog-client SplunkUniversalForwarder]#
[root@rsyslog-client SplunkUniversalForwarder]# cd local
[root@rsyslog-client local]# ls
README    default-mode.conf  inputs.conf  outputs.conf  server.conf
app.conf  health.conf        limits.conf  props.conf    web.conf
```

#mkdir local

#cp default/* local/

#cd local

*주요 환경 설정 파일 backup
(동일한 환경설정이 있는 경우 local 디렉터리 환경 설정 파일을 우선)

환경 설정 파일

- inputs.conf

  - 수집 대상 설정 (where)

- outputs.conf

  - 수집 로그 전달 받을 서버 설정 (where)

Splunk Universal Forwarder 환경 설정(P.206)

❶ 수집 대상 로그 파일 확인

❷ inputs.conf 파일 수정

❸ outputs.conf 파일 수정

❹ 로그 전송 여부 확인

❺ 로그 수신 여부 확인

❶ 수집 대상 로그 파일 확인

/var/log/httpd/access-log

```
[root@rsyslog-client httpd]# pwd
/var/log/httpd
[root@rsyslog-client httpd]# ls -l
합계 8
-rw-r--r--. 1 root root 1005  8월  7 12:57 access_log
-rw-r--r--. 1 root root 1602  8월  7 12:49 error_log
[root@rsyslog-client httpd]#
```

#tail access_log

```
[root@rsyslog-client httpd]# pwd
/var/log/httpd
[root@rsyslog-client httpd]# tail access_log
::1 - - [07/Aug/2022:13:08:09 +0900] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20100101 Firefox/60.0"
192.168.10.30 - - [07/Aug/2022:13:08:14 +0900] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20100101 Firef
192.168.10.30 - - [07/Aug/2022:13:08:14 +0900] "GET /favicon.ico HTTP/1.1" 404 209 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20
192.168.10.10 - - [07/Aug/2022:13:10:28 +0900] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/201001
192.168.10.10 - - [07/Aug/2022:13:10:28 +0900] "GET /favicon.ico HTTP/1.1" 404 209 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0)
5.0"
```

❷ inputs.conf 파일 수정

- /opt/splunkforwarder/etc/apps/SplunkUniversalForwarder/local/inputs.conf

열기(O)

inputs.conf
/opt/splunkforwarder/etc/apps/SplunkUniversalForwarder/local

```
#   Version 9.0.0.1
# these here just override and disable stuff that in system/default.

##############################
# Data thru parsingQueue always
##############################

[splunktcp]
route=has_key:tautology:parsingQueue;absent_key:tautology:parsingQueue

##############################
# Make sure these get forwarded
##############################

#[monitor://$SPLUNK_HOME/var/log/splunk/splunkd.log]
#_TCP_ROUTING = *

#[monitor://$SPLUNK_HOME/var/log/splunk/metrics.log]
#_TCP_ROUTING = *
#index = _internal

[monitor:///var/log/httpd/access_log]
disable =0
index=centhttpd
```

❸ outputs.conf 파일 수정

- /opt/splunkforwarder/etc/apps/SplunkUniversalForwarder/local/outputs.conf

열기(O) ▾ | outputs.conf | 저장(S)
/opt/splunkforwarder/etc/apps/SplunkUniversalForwarder/local

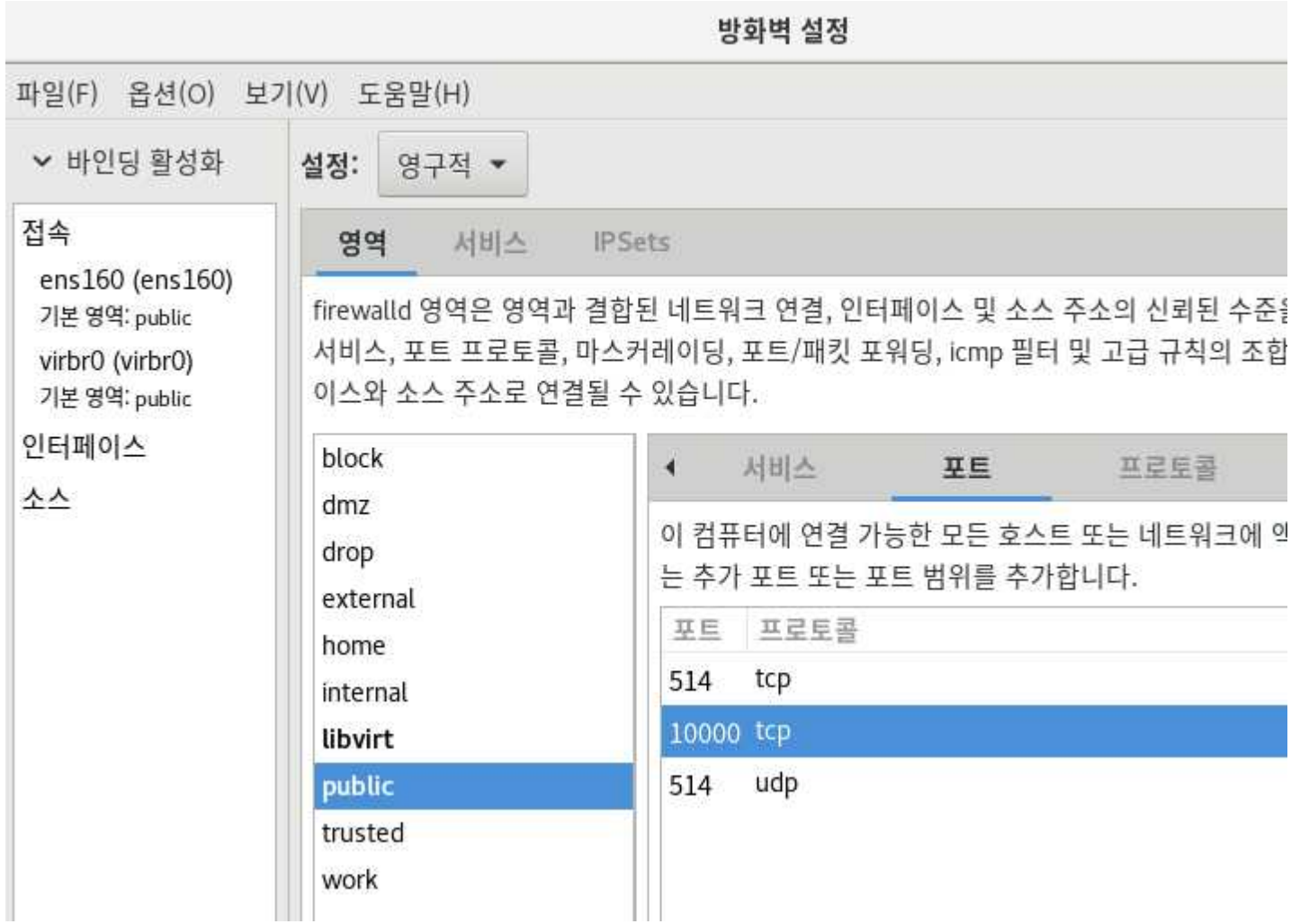
```
#   Version 9.0.0.1
[tcpout]
forwardedindex.0.whitelist = .*
forwardedindex.1.blacklist = _.*
forwardedindex.2.whitelist = (_audit|_introspection|_internal|_telemetry|_configtracker)
forwardedindex.filter.disable = false

defaultGroup=default-autolb-group

[tcpout:default-autolb-group]
server=192.168.10.10:10000

[tcpout-server://192.168.10.10:10000]
```

#cd /opt/splunkforwarder/bin

#./splunk restart

```
[root@rsyslog-client bin]# pwd
/opt/splunkforwarder/bin
[root@rsyslog-client bin]# ls
2to3-3.7     classify                genWebCert.sh  pip3         pripalpng    pydoc3        splunk
S3benchmark  copyright.txt           idle3          pip3.7       pripamtopng  pydoc3.7      splunkd
btool        easy_install-3.7        idle3.7        prichunkpng  pripnglsch   scripts       splunkmon
btprobe      genRootCA.sh            openssl        priforgepng  pripngtopam  setSplunkEnv  wheel
bzip2        genSignedServerCert.sh  pid_check.sh   prigreypng   priweavepng  slim
[root@rsyslog-client bin]#
[root@rsyslog-client bin]# ./splunk restart
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk /opt/splunkforwarder"
Stopping splunkd...
Shutting down.  Please wait, as this may take a few minutes.
                                                           [  OK  ]
Stopping splunk helpers...
                                                           [  OK  ]
Done.
splunkd.pid doesn't exist...
```

```
[root@rsyslog-client bin]# netstat -an | grep 10000
tcp        0      0 192.168.10.30:34790     192.168.10.10:10000     TIME_WAIT
tcp        0      0 192.168.10.30:34788     192.168.10.10:10000     TIME_WAIT
tcp        0      0 192.168.10.30:34792     192.168.10.10:10000     ESTABLISHED
[root@rsyslog-client bin]#
```


방화벽 설정
파일(F) 옵션(O) 보기(V) 도움말(H)
바인딩 활성화
설정: 영구적
접속
ens160 (ens160)
기본 영역: public
virbr0 (virbr0)
기본 영역: public
인터페이스
소스
영역 서비스 IPSets
firewalld 영역은 영역과 결합된 네트워크 연결, 인터페이스 및 소스 주소의 신뢰된 수준
서비스, 포트 프로토콜, 마스커레이딩, 포트/패킷 포워딩, icmp 필터 및 고급 규칙의 조합
이스와 소스 주소로 연결될 수 있습니다.
block
dmz
drop
external
home
internal
libvirt
public
trusted
work
서비스 포트 프로토콜
이 컴퓨터에 연결 가능한 모든 호스트 또는 네트워크에
는 추가 포트 또는 포트 범위를 추가합니다.
포트 프로토콜
514 tcp
10000 tcp
514 udp

splunk>enterprise Apps ▾ Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find

Indexes

New Index

A repository for data in Splunk Enterprise. Indexes reside in flat files on the Splunk Enterprise instance known as the indexer. Learn more

1 Indexes centhttpd X 20 per page ▾

Name	Actions	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path	Status
centhttpd	Edit Delete Disable	Events	search	1 MB	500 GB	8	a minute ago	a few seconds ago	$SPLUNK_DB/centhttpd/db	N/A	✓ Enabled

splunk>enterprise Apps ▾ Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find

Search Analytics Datasets Reports Alerts Dashboards Search & R

New Search

Save As ▾ Create Table View

index=centhttpd Last 24 hours ▾

✓ **8 events** (8/6/22 2:00:00.000 PM to 8/7/22 2:13:16.000 PM) No Event Sampling ▾ Job ▾ Smart

Events (8) Patterns Statistics Visualization

Format Timeline ▾ − Zoom Out + Zoom to Selection × Deselect 1 hour

List ▾ Format 20 Per Page ▾

< Hide Fields All Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

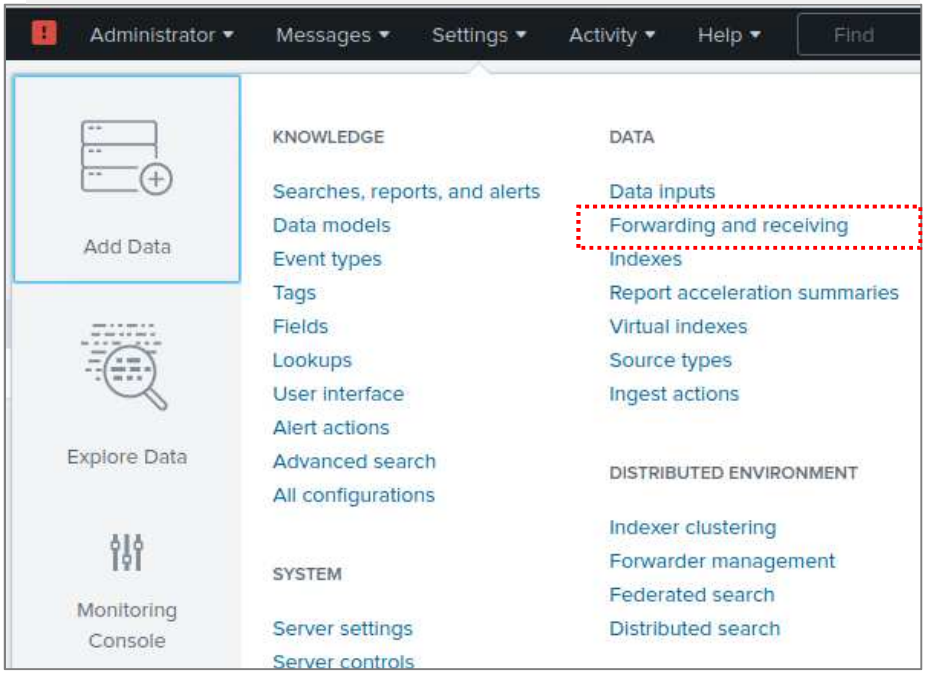
INTERESTING FIELDS
date_hour 1

i	Time	Event
>	8/7/22 2:10:52.000 PM	192.168.10.20 - - [07/Aug/2022:14:10:52 +0900] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0" host = rsyslog-client source = /var/log/httpd/access.log sourcetype = access-too_small
>	8/7/22 2:10:52.000 PM	192.168.10.20 - - [07/Aug/2022:14:10:52 +0900] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0" host = rsyslog-client source = /var/log/httpd/access.log sourcetype = access-too_small
>	8/7/22 2:10:52.000 PM	192.168.10.20 - - [07/Aug/2022:14:10:52 +0900] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0" host = rsyslog-client source = /var/log/httpd/access.log sourcetype = access-too_small

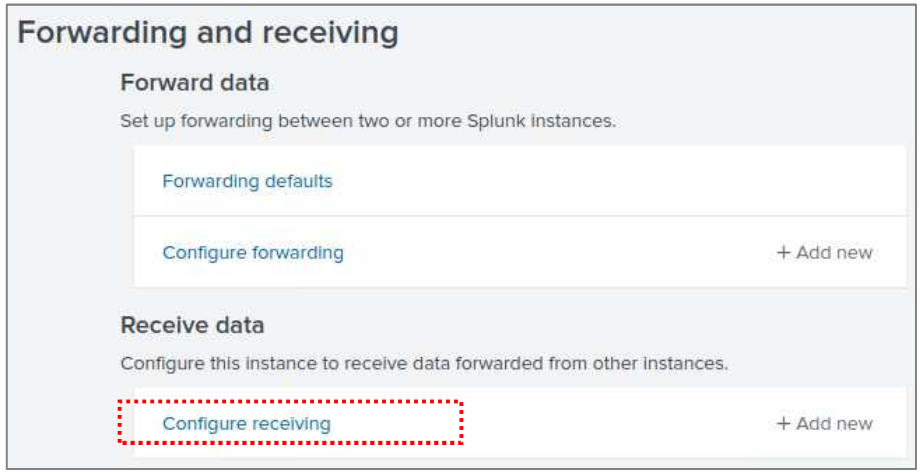
3) [Splunk Server] Forwarder 수집 환경 설정 (P.205)

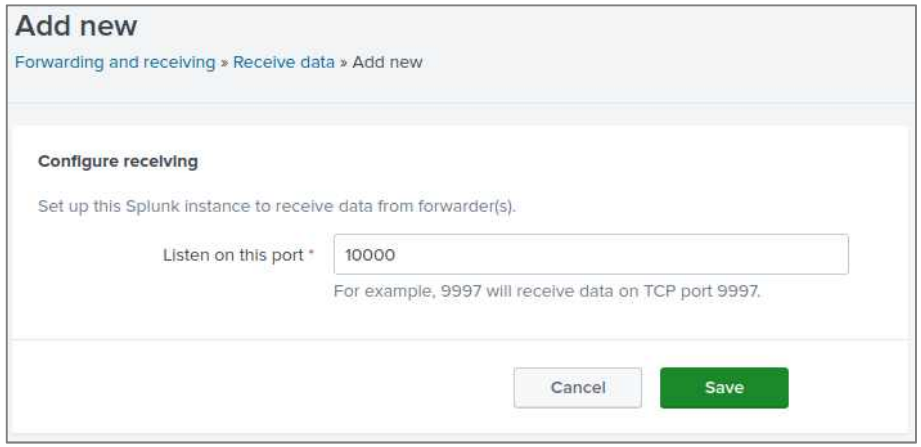
❶ Forward로부터 받을 포트 설정

Setting > Data
> Forwarding and receiving
> Receive data > Configure receiving
> Add new



** Splunk Forwarder를 위한
Splunk Server 설정





❷ Receiving port open 설정(방화벽설정)

```
#ufw allow 10000/tcp

#ufw status

#netstat –an | grep 10000
```

```
root@splunk-server:~# ufw  allow 10000/tcp
규칙이 추가되었습니다
규칙이 추가되었습니다 (v6)
root@splunk-server:~# ufw status
상태: 활성

목적                       동작         출발
--                       --         --
123/udp                  ALLOW      Anywhere
514/tcp                  ALLOW      Anywhere
514/udp                  ALLOW      Anywhere
22/tcp                   ALLOW      Anywhere
10000/tcp                ALLOW      Anywhere
123/udp (v6)             ALLOW      Anywhere (v6)
514/tcp (v6)             ALLOW      Anywhere (v6)
514/udp (v6)             ALLOW      Anywhere (v6)
22/tcp (v6)              ALLOW      Anywhere (v6)
10000/tcp (v6)           ALLOW      Anywhere (v6)
root@splunk-server:~# netstat -an | grep 10000
tcp        0      0 0.0.0.0:10000           0.0.0.0:*               LISTEN
root@splunk-server:~#
```

4) [Win10] Splunk Universal Forwarder Installation

❶ Splunk Universal Forwarder 설치

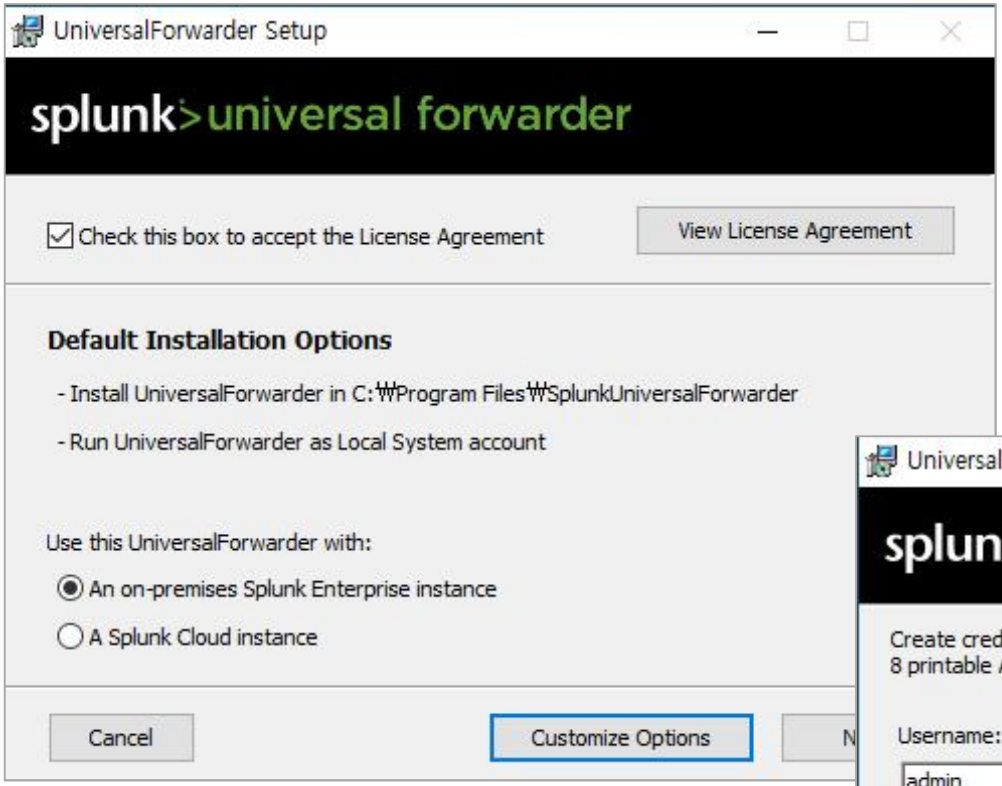
Splunk Universal Forwarder 9.0.0.1

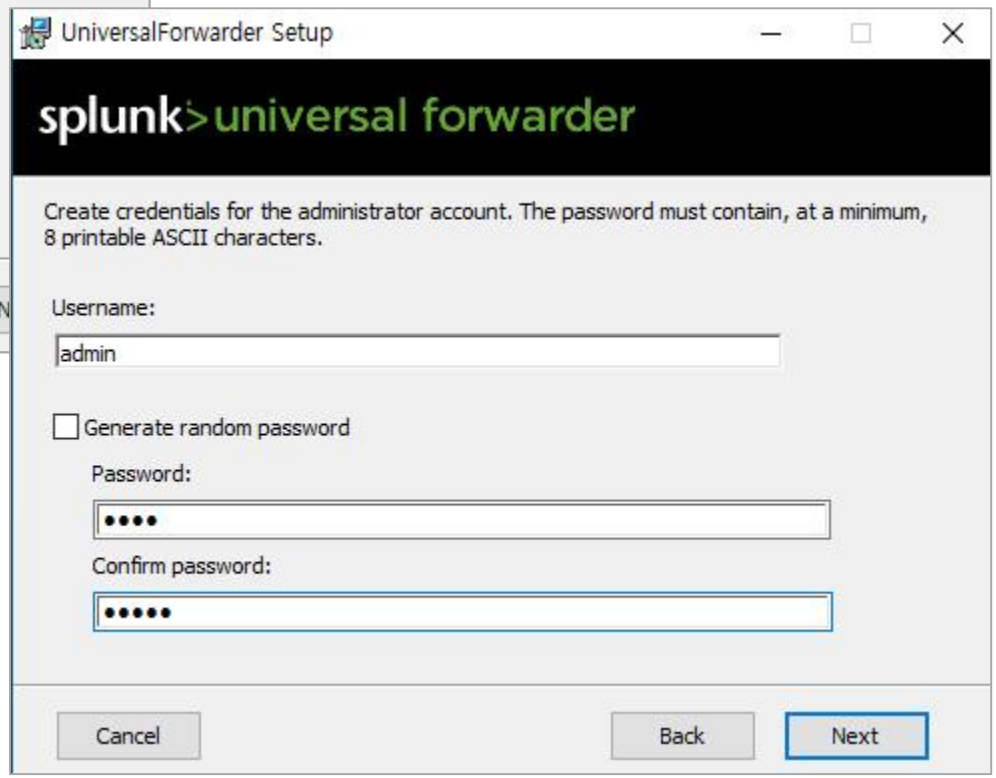
Universal Forwarders provide reliable, secure data collection from remote sources and forward that data into Splunk software for indexing and consolidation. They can scale to tens of thousands of remote systems, collecting terabytes of data.

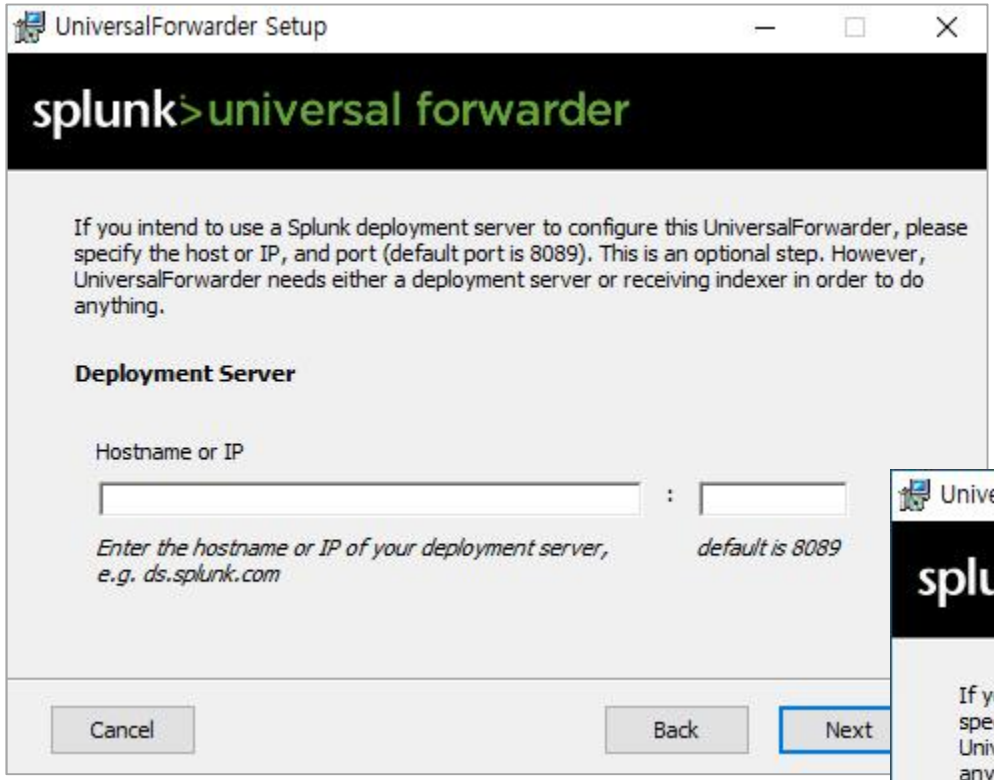
Choose Your Installation Package

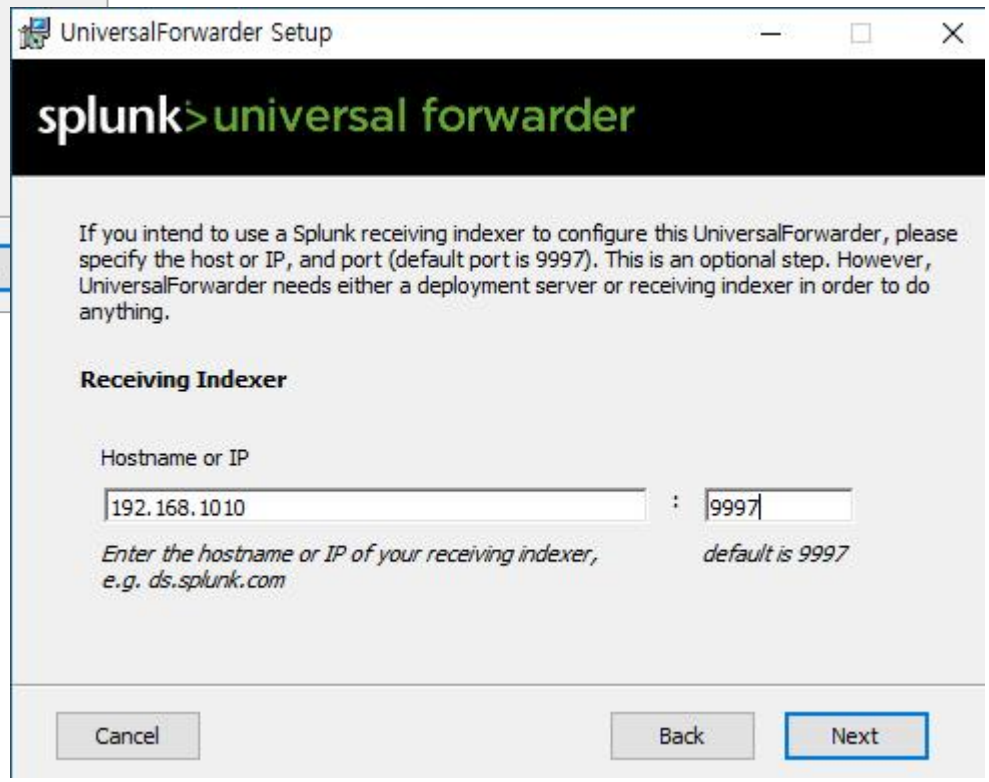
Windows | Linux | Mac OS | Free BSD | Solaris | AIX

64-bit	Windows 10 , Windows 11 Windows Server 2012, 2012 R2, 2016, 2019, 2022	.msi	75.41 MB	Download Now
32-bit	Windows 10	.msi	62.49 MB	Download Now


UniversalForwarder Setup
splunk>universal forwarder
Check this box to accept the License Agreement
View License Agreement
Default Installation Options
- Install UniversalForwarder in C:₩Program Files₩SplunkUniversalForwarder
- Run UniversalForwarder as Local System account
Use this UniversalForwarder with:
An on-premises Splunk Enterprise instance
A Splunk Cloud instance
Cancel
Customize Options


UniversalForwarder Setup
splunk>universal forwarder
Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.
Username:
admin
Generate random password
Password:
Confirm password:
Cancel
Back
Next


UniversalForwarder Setup
splunk>universal forwarder
If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.
Deployment Server
Hostname or IP
:
Enter the hostname or IP of your deployment server, e.g. ds.splunk.com
default is 8089
Cancel
Back
Next


UniversalForwarder Setup
splunk>universal forwarder
If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.
Receiving Indexer
Hostname or IP
192.168.1010
:
9997
Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com
default is 9997
Cancel
Back
Next

Program Files › SplunkUniversalForwarder › etc › apps › SplunkUniversalForwarder › local

이름	수정한 날짜	유형	크기
app.conf	2022-07-30 오후 4:13	CONF 파일	
inputs.conf	2022-07-30 오후 4:35	CONF 파일	

inputs.txt - Windows 메모장

파일(F) 편집(E) 서식(O) 보기(V) 도움말(H)

```
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
checkpointInterval=5
current_only=0
disabled=0
start_from=oldest
index=sysmon

[WinEventLog://Application]
checkpointInterval=5
current_only=0
disabled=0
start_from=oldest
index=sysmon

[WinEventLog://Security]
checkpointInterval=5
current_only=0
disabled=0
start_from=oldest
index=sysmon

[WinEventLog://system]
checkpointInterval=5
current_only=0
disabled=0
start_from=oldest
index=sysmon
```

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
checkpointInterval=5
current_only=0
disabled=0
start_from=oldest
index=sysmon

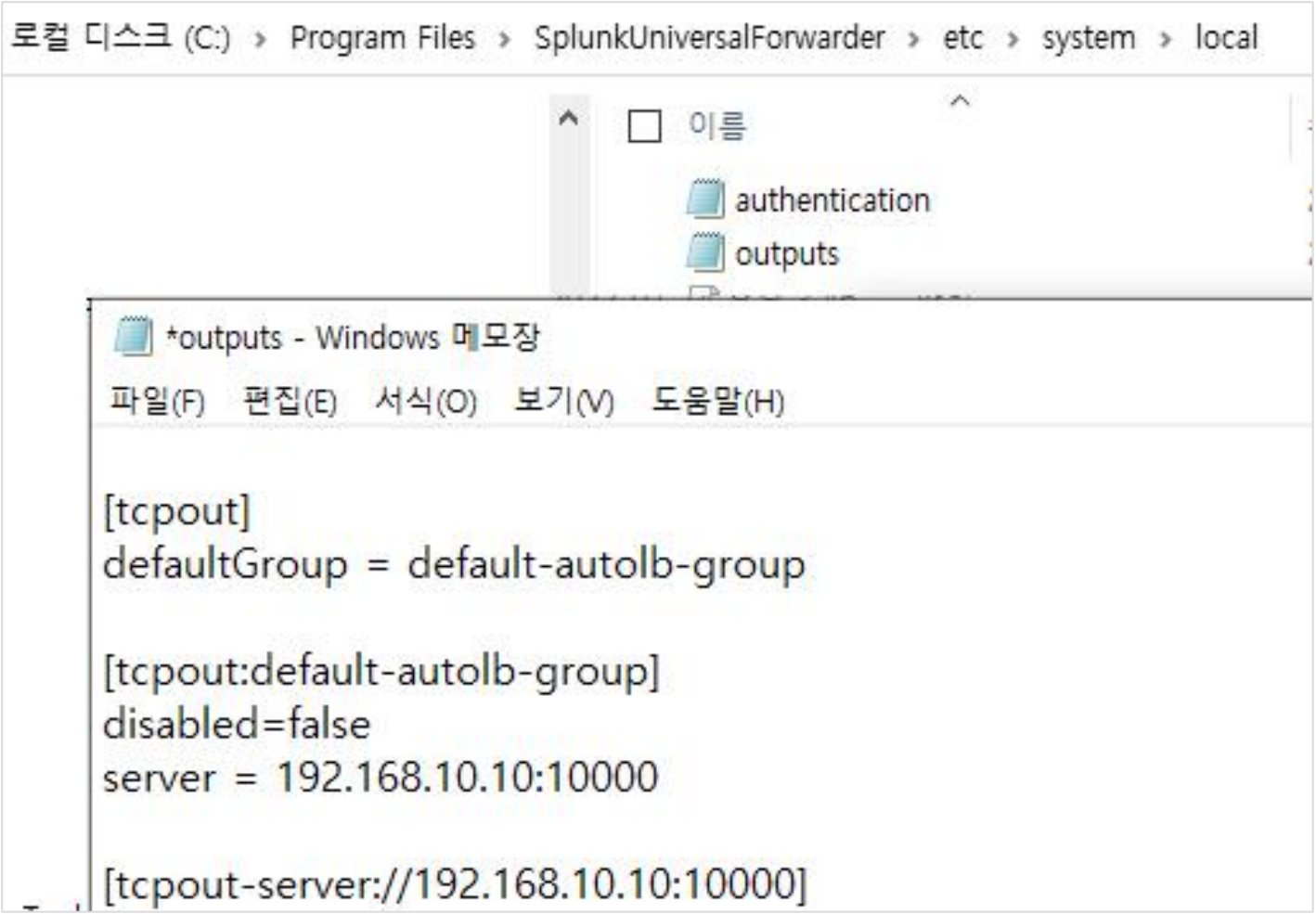
[WinEventLog://Application]
[WinEventLog://Security]
[WinEventLog://system]

❷ Forwarder 환경설정

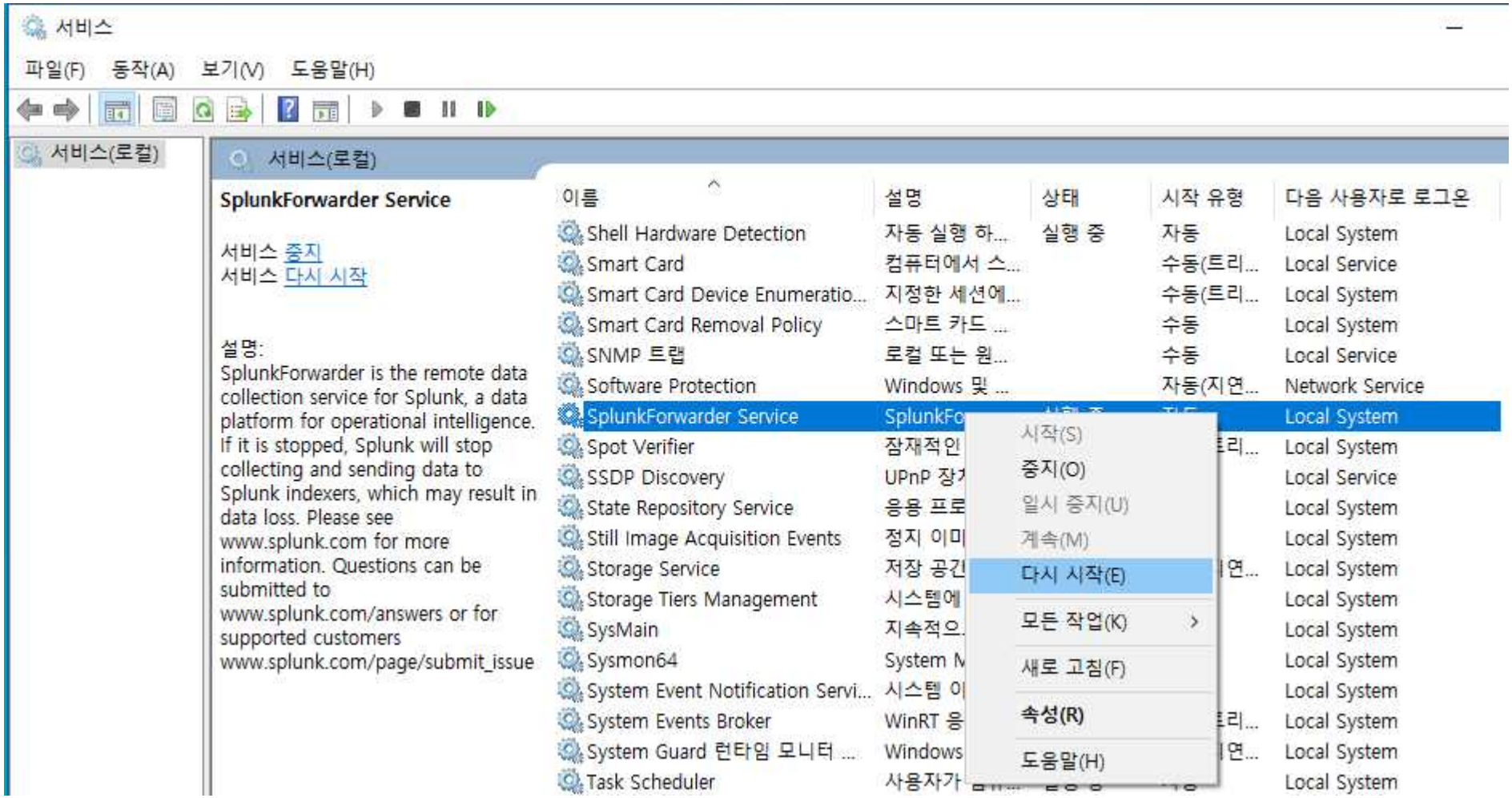
* inputs.conf 파일 설정

❷ Forwarder 환경설정

* outputs.conf 파일 설정



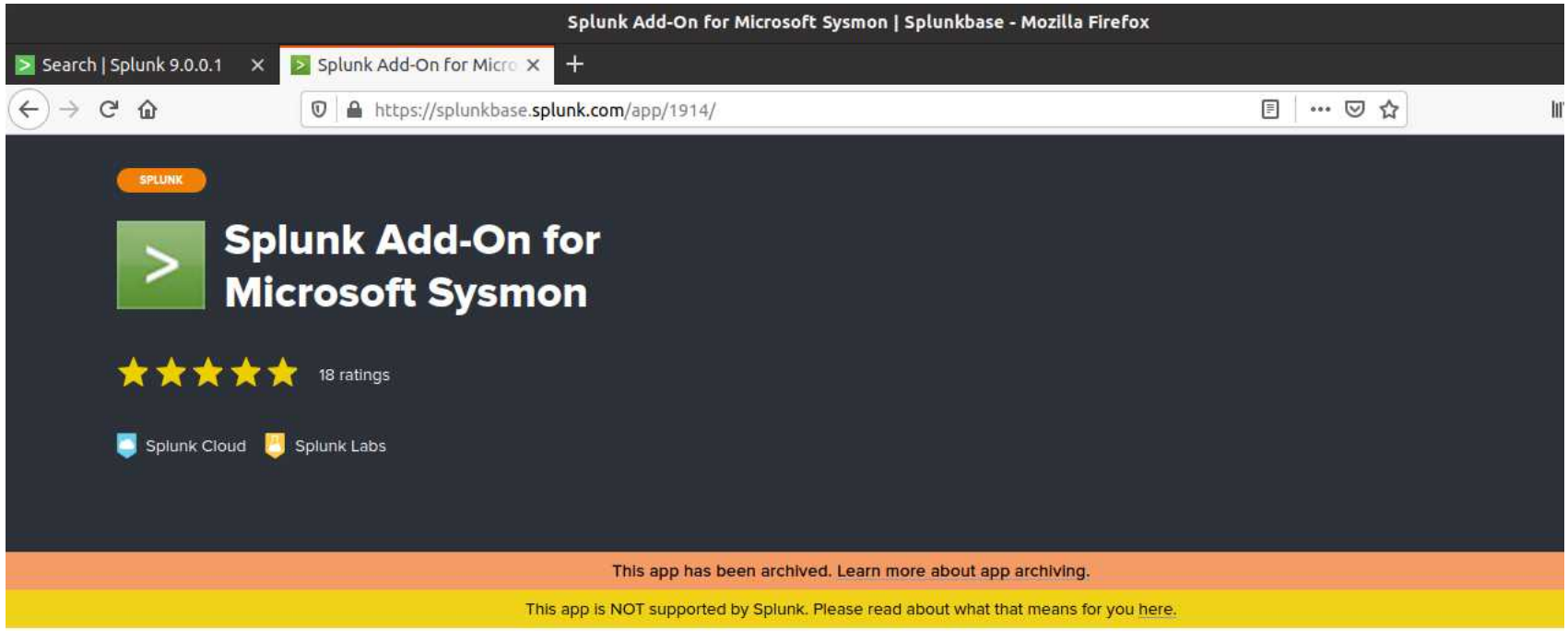
❸ Forwarder 서비스 활성화

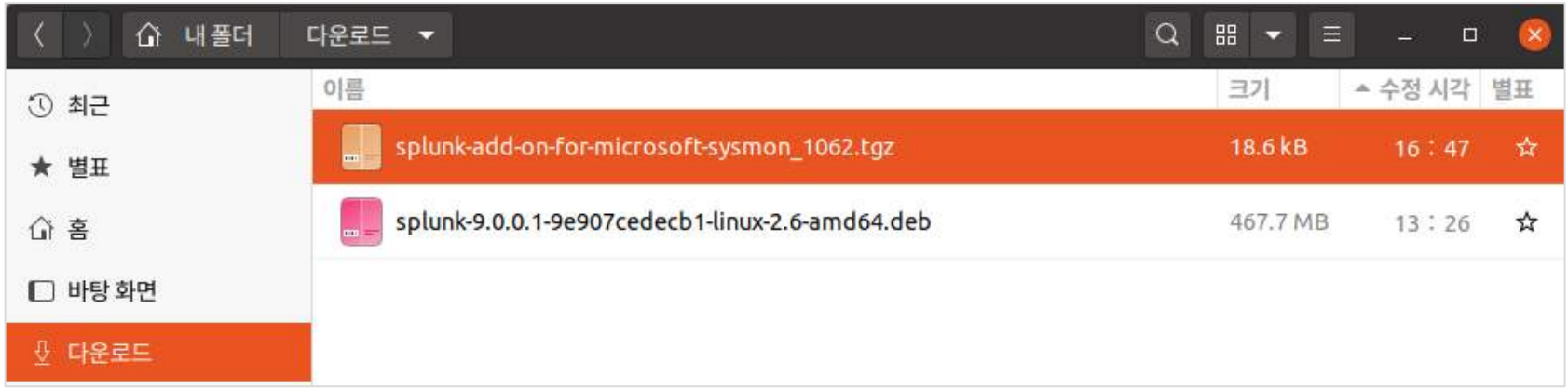


5) [Splunk Server] Sysmon 전용 프로그램 설치

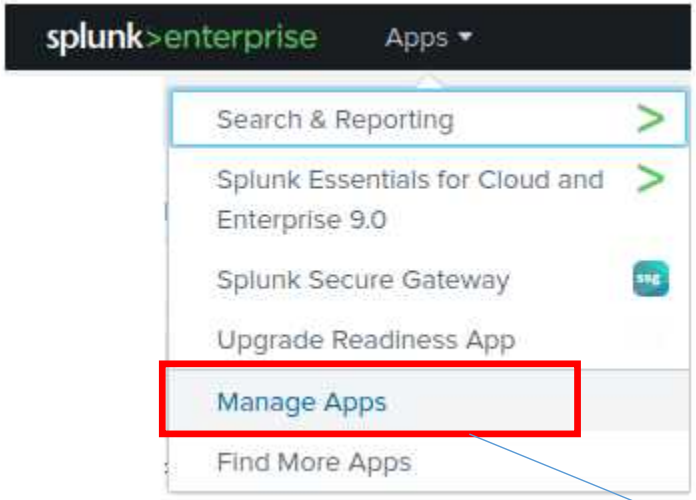
❶ Splunk Add-on 프로그램 다운로드

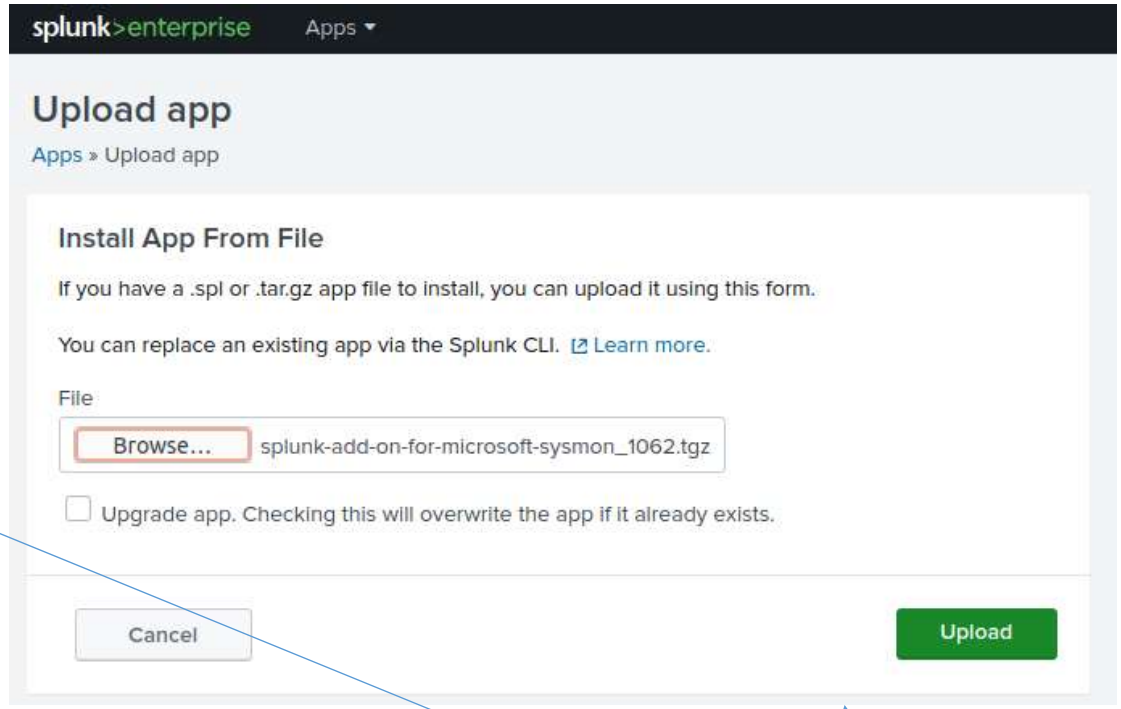
https://splunkbase.splunk.com/app/1914

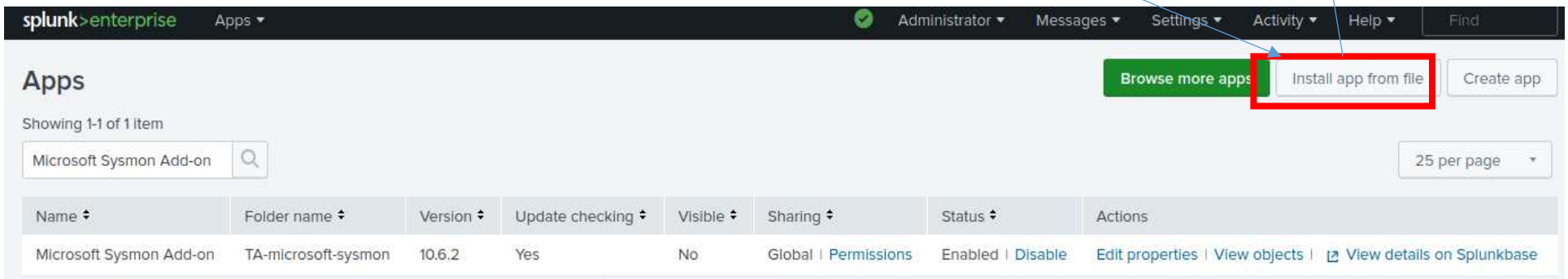



내 폴더
다운로드
최근
별표
홈
바탕 화면
다운로드
이름
크기
수정 시각
별표
splunk-add-on-for-microsoft-sysmon_1062.tgz
18.6 kB
16 : 47
splunk-9.0.0.1-9e907cedecb1-linux-2.6-amd64.deb
467.7 MB
13 : 26

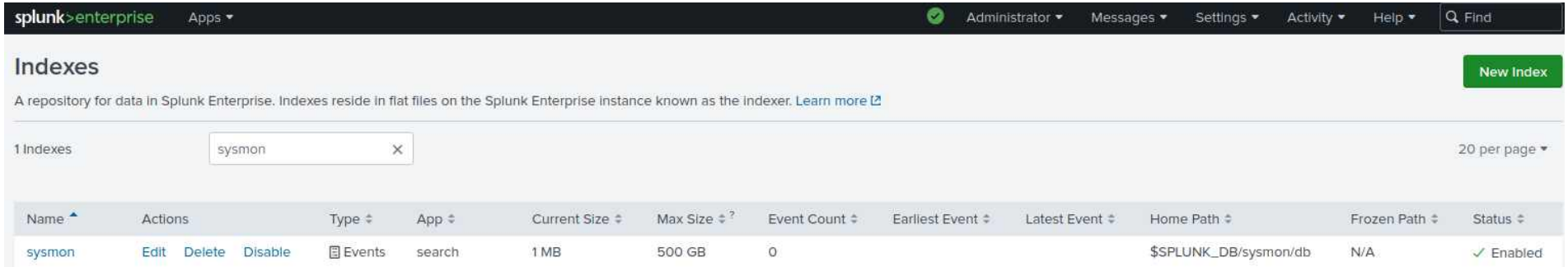
❷ Splunk Add-on 프로그램 설치





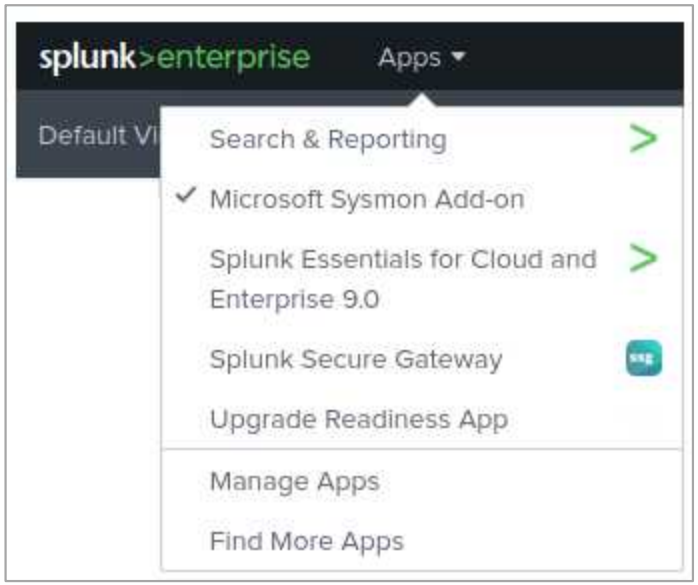


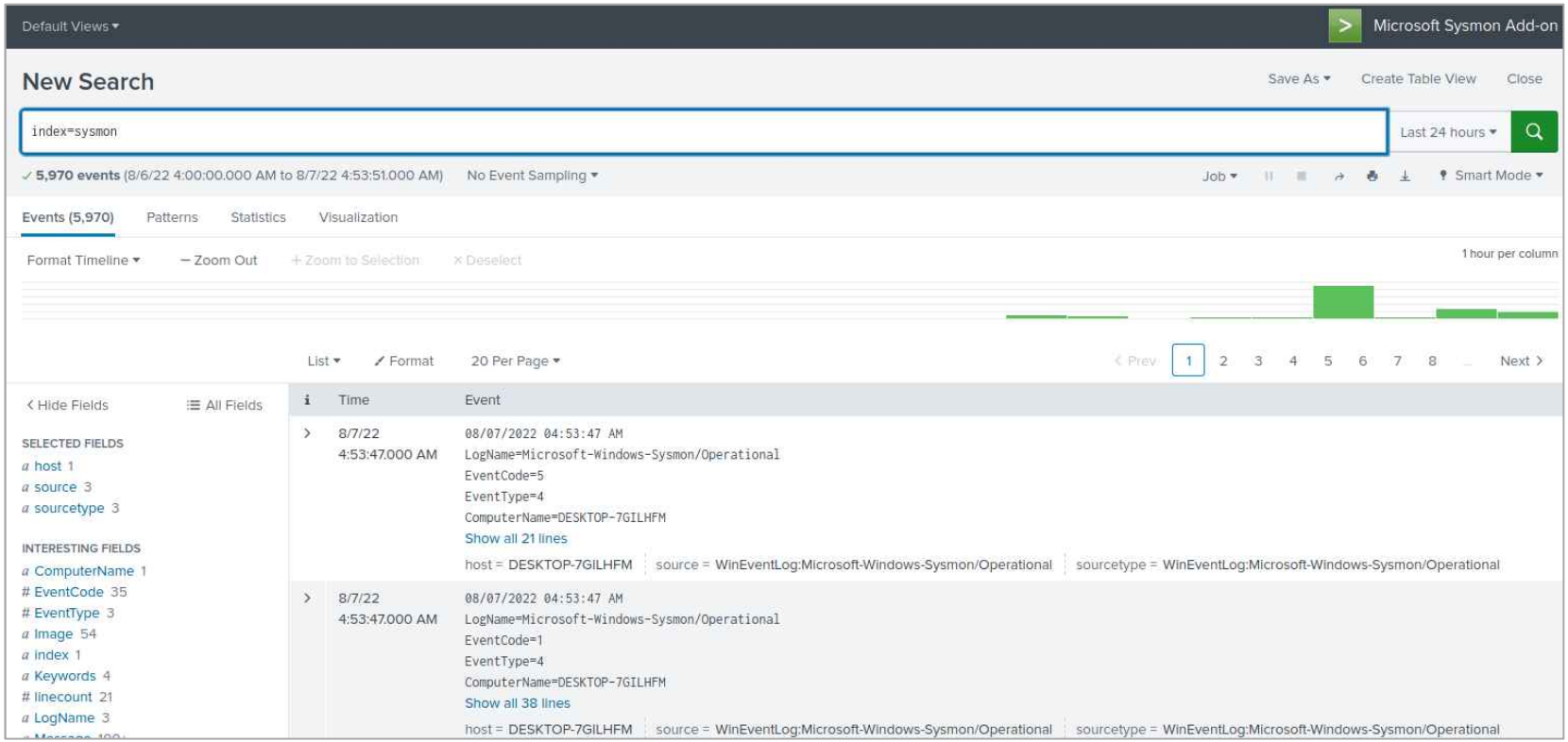
❸ Index 생성



❹ Splunk 재시작

Settings > SYSTEM > Server Controls


splunk>enterprise
Apps
Default Vi
Search & Reporting
Microsoft Sysmon Add-on
Splunk Essentials for Cloud and Enterprise 9.0
Splunk Secure Gateway
Upgrade Readiness App
Manage Apps
Find More Apps


Default Views
Microsoft Sysmon Add-on
New Search
Save As
Create Table View
Close
index=sysmon
Last 24 hours
5,970 events (8/6/22 4:00:00.000 AM to 8/7/22 4:53:51.000 AM)
No Event Sampling
Job
Smart Mode
Events (5,970)
Patterns
Statistics
Visualization
Format Timeline
Zoom Out
Zoom to Selection
Deselect
1 hour per column
List
Format
20 Per Page
Prev
1 2 3 4 5 6 7 8
Next
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 3
sourcetype 3
INTERESTING FIELDS
ComputerName 1
EventCode 35
EventType 3
Image 54
index 1
Keywords 4
linecount 21
LogName 3
Time
Event
8/7/22 4:53:47.000 AM
08/07/2022 04:53:47 AM
LogName=Microsoft-Windows-Sysmon/Operational
EventCode=5
EventType=4
ComputerName=DESKTOP-7GILHFM
Show all 21 lines
host = DESKTOP-7GILHFM
source = WinEventLog:Microsoft-Windows-Sysmon/Operational
sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
8/7/22 4:53:47.000 AM
08/07/2022 04:53:47 AM
LogName=Microsoft-Windows-Sysmon/Operational
EventCode=1
EventType=4
ComputerName=DESKTOP-7GILHFM
Show all 38 lines
host = DESKTOP-7GILHFM
source = WinEventLog:Microsoft-Windows-Sysmon/Operational
sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational